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FUNCTIONAL ANALYSIS

Complete Study Notes — Days 5 to 10

Normed Spaces · Banach Spaces · Inner Product & Hilbert Spaces

Hahn-Banach Theorem · Uniform Boundedness · Open Mapping

Closed Graph · Compact Operators · Spectral Theory

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AJKPSC Mathematics Test Preparation

Detailed Notes + All Key Theorems + Full Proofs + **200 Solved MCQs (40 per day)**

Day 5	**Day 6**	**Day 7**	**Day 8**	**Day 9–10**
Normed & Banach Spaces	Inner Product & Hilbert Spaces	Hahn-Banach Theorem ★	The Big Three Theorems ★★	Operators & Revision

Contents

[DAY] DAY 5 — Normed Spaces & Banach Spaces

Topics: Normed linear spaces, convergence, completeness, Banach spaces (ℓ^p, $C[a,b]$, L^p), equivalent norms, finite-dimensional normed spaces.

1. Normed Linear Spaces

[DEF] Normed Linear Space

Let X be a vector space over $\mathbb{K}$ ($= \mathbb{R}$ or $\mathbb{C}$). A **norm** on X is a function $\|\cdot\| : X \to \mathbb{R}_{\geq 0}$ satisfying:

(N1) $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$ *(positive definiteness)*

(N2) $\|\alpha x\| = |\alpha|\,\|x\|$ for all $\alpha \in \mathbb{K}, x \in X$ *(absolute homogeneity)*

(N3) $\|x + y\| \leq \|x\| + \|y\|$ *(triangle inequality)*

The pair $(X, \|\cdot\|)$ is called a **normed linear space (NLS)**.

[EX] Standard Normed Spaces

- $(\mathbb{R}^n, \|\cdot\|_p)$: $\|x\|_p = \left(\sum |x_i|^p\right)^{1/p}$ for $1 \leq p < \infty$; $\|x\|_\infty = \max_i |x_i|$.
- $(C[a,b], \|\cdot\|_\infty)$: $\|f\|_\infty = \max_{t\in[a,b]} |f(t)|$ (sup norm).
- $(\ell^p, \|\cdot\|_p)$: $\ell^p = \{(x_n) : \sum |x_n|^p < \infty\}$, $\|x\|_p = \left(\sum |x_n|^p\right)^{1/p}$.
- $(\ell^\infty, \|\cdot\|_\infty)$: bounded sequences with $\|x\|_\infty = \sup_n |x_n|$.
- $(L^p[a,b], \|\cdot\|_p)$: $\|f\|_p = \left(\int_a^b |f|^p\right)^{1/p}$.

1.1. Convergence and Completeness

[DEF] Convergence in NLS

In a normed space $(X, \|\cdot\|)$:

- $x_n \to x$ (converges to x) if $\|x_n - x\| \to 0$.
- (x_n) is **Cauchy** if $\|x_m - x_n\| \to 0$ as $m, n \to \infty$.
- $\sum x_n$ **converges** if the partial sums $s_N = \sum_{n=1}^N x_n$ converge.

Every normed space is a metric space via $d(x, y) = \|x - y\|$.

1.2. Banach Spaces

[DEF] Banach Space

A normed linear space $(X, \|\cdot\|)$ is a **Banach space** if it is *complete*: every Cauchy sequence converges to an element of X.

[THM] Important Banach Spaces

(i) $\mathbb{R}^n$ and $\mathbb{C}^n$ with any norm are Banach spaces.

(ii) $(C[a,b], \|\cdot\|_\infty)$ is a Banach space.

(iii) $(\ell^p, \|\cdot\|_p)$ is a Banach space for $1 \leq p \leq \infty$.

(iv) $(L^p[a,b], \|\cdot\|_p)$ is a Banach space for $1 \le p \le \infty$.

(v) Any closed subspace of a Banach space is a Banach space.

[THM] Series Criterion for Completeness

A normed space X is a Banach space $\iff$ every **absolutely convergent** series converges in X:

$$\sum_{n=1}^{\infty} \|x_n\| < \infty \implies \sum_{n=1}^{\infty} x_n \text{ converges in } X.$$

1.3. Equivalent Norms

[DEF] Equivalent Norms

Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on X are **equivalent** if there exist constants $c, C > 0$ such that:

$$c\,\|x\|_a \;\le\; \|x\|_b \;\le\; C\,\|x\|_a \quad \forall\, x \in X.$$

Equivalent norms define the same topology (same open sets, same convergent sequences).

[THM] Finite-Dimensional Normed Spaces

(i) All norms on a **finite-dimensional** vector space are equivalent.

(ii) Every finite-dimensional normed space is a **Banach space**.

(iii) Every finite-dimensional normed space is **locally compact**.

(iv) A normed space is finite-dimensional $\iff$ its closed unit ball is compact (Riesz's theorem).

[PROOF] Proof: $C[a,b]$ with Sup Norm is Complete (Banach)

Let (f_n) be Cauchy in $(C[a,b], \|\cdot\|_\infty)$. Then for each $t \in [a,b]$: $|f_m(t) - f_n(t)| \le \|f_m - f_n\|_\infty \to 0$. So $(f_n(t))$ is Cauchy in $\mathbb{R}$, hence converges; define $f(t) = \lim_n f_n(t)$. The convergence is *uniform* (since $\|f_n - f_m\|_\infty \to 0$ uniformly). Uniform limit of continuous functions is continuous, so $f \in C[a,b]$. Finally $\|f_n - f\|_\infty \to 0$. So $(C[a,b], \|\cdot\|_\infty)$ is complete. □

Day 5 MCQs — Normed Spaces & Banach Spaces

Q1

A **norm** on a vector space X must satisfy:

(A) Positivity, linearity, and boundedness

(B) Positive definiteness, absolute homogeneity, and triangle inequality

(C) Symmetry, additivity, and non-negativity

(D) Continuity, linearity, and completeness

Answer: (B)

Explanation: *The three norm axioms are: (N1) $\|x\| \ge 0$ and $\|x\| = 0 \iff x = 0$; (N2) $\|\alpha x\| = |\alpha|\|x\|$; (N3) $\|x+y\| \le \|x\| + \|y\|$. "Linearity" in (A) confuses norm with linear map; a norm is positively homogeneous, not linear.*

Q2

A **Banach space** is a normed linear space that is:
(A) Finite-dimensional
(B) Complete (every Cauchy sequence converges)
(C) Compact
(D) Separable

Answer: (B)

Explanation: *Banach space = complete normed linear space. Completeness means every Cauchy sequence has its limit inside the space. $\mathbb{R}^n$, $C[a,b]$ (with sup norm), ℓ^p are all Banach spaces; $\mathbb{Q}$ with $|\cdot|$ is a normed space but NOT Banach (not complete).*

Q3

Which of the following is **NOT** a Banach space?
(A) $(C[0,1], \|\cdot\|_\infty)$
(B) $(\ell^2, \|\cdot\|_2)$
(C) $(C[0,1], \|\cdot\|_1)$ where $\|f\|_1 = \int_0^1 |f(t)|dt$
(D) $(\mathbb{R}^n, \text{Euclidean norm})$

Answer: (C)

Explanation: *$(C[0,1], \|\cdot\|_1)$ is NOT complete. Counterexample: $f_n(t) = t^n$ is Cauchy in L^1 norm but the limit $f(t) = \mathbf{1}_{\{1\}}(t)$ (pointwise) is not continuous. The sup norm makes $C[0,1]$ Banach, but the L^1 norm does not.*

Q4

The ℓ^p **space** $(1 \le p < \infty)$ consists of sequences (x_n) with:
(A) $\sup_n |x_n| < \infty$
(B) $\sum_{n=1}^\infty |x_n|^p < \infty$
(C) $\lim_{n\to\infty} x_n = 0$
(D) $\sum_{n=1}^\infty |x_n| < \infty$

Answer: (B)

Explanation: *$\ell^p = \{(x_n) : \sum |x_n|^p < \infty\}$ with norm $\|(x_n)\|_p = (\sum |x_n|^p)^{1/p}$. (A) is ℓ^∞; (C) is c_0 (sequences vanishing at infinity); (D) is ℓ^1 specifically.*

Q5

In a normed space, $x_n \to x$ means:
(A) $\|x_n\| \to \|x\|$
(B) $\|x_n - x\| \to 0$
(C) $x_n = x$ eventually
(D) $\|x_n + x\| \to 0$

Answer: (B)

Explanation: *Convergence in a normed space: $x_n \to x \iff \|x_n - x\| \to 0$. Note (A) is weaker (norm convergence does not imply strong convergence). (C) is "eventually constant," stronger. The norm $\|\cdot\|$ is continuous so $x_n \to x \Rightarrow \|x_n\| \to \|x\|$, but not vice versa.*

Q6

Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on X are **equivalent** if:
(A) They are equal: $\|x\|_a = \|x\|_b$ for all x
(B) $\exists\, c, C > 0$ such that $c\|x\|_a \le \|x\|_b \le C\|x\|_a$ for all x
(C) One is a scalar multiple of the other
(D) They produce the same Cauchy sequences only

Answer: (B)

Explanation: *Equivalent norms: sandwiched by positive constants. Equivalent norms define the same topology (same open sets, same convergent and Cauchy sequences). In finite dimensions, ALL norms are equivalent.*

Q7

All norms on a **finite-dimensional** vector space are:
(A) Equal
(B) Equivalent
(C) Inequivalent
(D) Only equivalent if the space is over $\mathbb{R}$

Answer: (B)

Explanation: *Theorem: on a finite-dimensional vector space, all norms are equivalent. This implies all finite-dimensional normed spaces are complete (Banach), locally compact, and have the same convergent sequences regardless of norm chosen.*

Q8

Riesz's theorem: a normed space is finite-dimensional $\iff$:

(A) It is separable

(B) Its closed unit ball $\{x : \|x\| \le 1\}$ is compact

(C) Every bounded sequence converges

(D) It is a Hilbert space

Answer: (B)

Explanation: *Riesz's theorem (1918): X finite-dimensional $\iff$ the closed unit ball $B[0,1]$ is compact. In infinite dimensions, the unit ball is never compact. This is the key distinction between finite and infinite-dimensional normed spaces.*

Q9

A normed space X is Banach $\iff$ every **absolutely convergent** series converges, i.e.:

(A) $\sum \|x_n\| < \infty \Rightarrow \sum x_n$ diverges

(B) $\sum \|x_n\| < \infty \Rightarrow \sum x_n$ converges in X

(C) $\sum x_n$ converges$\Rightarrow \sum \|x_n\| < \infty$

(D) Every bounded sequence has a convergent subsequence

Answer: (B)

Explanation: *Series criterion: X Banach $\iff$ absolute convergence implies convergence. This is a very useful test. In $\mathbb{R}$: if $\sum |a_n| < \infty$ then $\sum a_n$ converges — this is exactly why $\mathbb{R}$ is complete/Banach.*

Q10

The **sup norm** on $C[a,b]$ is:

(A) $\|f\| = \int_a^b |f(t)|dt$

(B) $\|f\| = \max_{t\in[a,b]} |f(t)|$

(C) $\|f\| = \left(\int_a^b |f(t)|^2 dt\right)^{1/2}$

(D) $\|f\| = |f(a)| + |f(b)|$

Answer: (B)

Explanation: *The supremum (uniform) norm: $\|f\|_\infty = \max_{t\in[a,b]} |f(t)|$ (maximum exists since f continuous on compact $[a,b]$). This makes $C[a,b]$ a Banach space. The other norms in (A) and (C) exist but do NOT make $C[a,b]$ complete.*

Q11

Which inclusion holds between ℓ^p spaces for $1 \le p < q \le \infty$?

(A) $\ell^q \subseteq \ell^p$

(B) $\ell^p \subseteq \ell^q$

(C) $\ell^p = \ell^q$

(D) No inclusion holds

Answer: (B)

Explanation: *$\ell^p \subseteq \ell^q$ for $p < q$. If $\sum |x_n|^p < \infty$ and $|x_n| \le 1$ eventually, then $|x_n|^q \le |x_n|^p$, giving convergence. The smaller the p, the stronger the condition. So $\ell^1 \subseteq \ell^2 \subseteq \cdots \subseteq \ell^\infty$.*

Q12

A **closed subspace** of a Banach space is:

(A) Compact

(B) A Banach space

(C) Finite-dimensional

(D) Dense in the whole space

Answer: (B)

Explanation: *Closed subspace of Banach = Banach. Proof: A Cauchy sequence in a closed subspace Y converges in the Banach space X to some $x \in X$; since Y is closed, $x \in Y$. So Y is complete, hence Banach.*

Q13

The norm $\|f\|_1 = \int_0^1 |f(t)|dt$ on $C[0,1]$:

(A) Makes $C[0,1]$ a Banach space

(B) Is equivalent to the sup norm on $C[0,1]$

(C) Does NOT make $C[0,1]$ a Banach space (it is incomplete)

(D) Is not a valid norm

Answer: (C)

Explanation: *$(C[0,1], \|\cdot\|_1)$ is NOT complete. The sequence $f_n(t) = t^n$ satisfies $\|f_n - 0\|_1 = 1/(n+1) \to 0$ but the pointwise limit is $\mathbf{1}_{\{1\}}$, which is not in $C[0,1]$. The completion of $(C[0,1], \|\cdot\|_1)$ is $L^1[0,1]$.*

Q14

Hölder's inequality states: for $\frac{1}{p} + \frac{1}{q} = 1$, $p, q \ge 1$:

(A) $\|x+y\|_p \le \|x\|_p + \|y\|_p$

(B) $\sum |x_n y_n| \le \|x\|_p \|y\|_q$

(C) $\|xy\|_p \le \|x\|_p \|y\|_p$

(D) $\|x\|_p \leq \|x\|_q$ for all $p \leq q$

Answer: (B)

Explanation: *Hölder's inequality: $\sum |x_n y_n| \leq \|x\|_p \|y\|_q$ where $1/p + 1/q = 1$ (conjugate exponents). Special case $p = q = 2$: Cauchy-Schwarz $\sum |x_n y_n| \leq \|x\|_2 \|y\|_2$. (A) is the triangle inequality (Minkowski's inequality).*

Q15

Every normed space is a:

(A) Hilbert space
(B) Banach space
(C) Metric space
(D) Compact space

Answer: (C)

Explanation: *Every normed space is a metric space via $d(x, y) = \|x - y\|$. Not every NLS is Banach (needs completeness) or Hilbert (needs inner product). Not every NLS is compact ($\mathbb{R}^n$ for $n \geq 1$ is not compact).*

Q16

The **dual exponent** of $p = 3$ (i.e., q with $1/p + 1/q = 1$) is:

(A) $q = 3$
(B) $q = 3/2$
(C) $q = 2/3$
(D) $q = 6$

Answer: (B)

Explanation: *$\frac{1}{3} + \frac{1}{q} = 1 \Rightarrow \frac{1}{q} = \frac{2}{3} \Rightarrow q = \frac{3}{2}$. The conjugate (dual) exponent of p is $q = p/(p-1)$. Special pairs: $(1, \infty)$, $(2, 2)$, $(4, 4/3)$.*

Q17

A sequence (x_n) in a normed space is **Cauchy** if:

(A) $\|x_n\| \to 0$
(B) $\|x_{n+1} - x_n\| \to 0$
(C) $\forall \varepsilon > 0$, $\exists N$: $m, n \geq N \Rightarrow \|x_m - x_n\| < \varepsilon$
(D) The sequence is bounded

Answer: (C)

Explanation: *Correct Cauchy definition: all terms become close to each other, not just consecutive ones. (B) is weaker (harmonic series is an example where $|a_{n+1} - a_n| \to 0$ but the series diverges). Cauchy is the key concept for completeness.*

Q18

In a Banach space, every Cauchy sequence:

(A) Is bounded but may not converge
(B) Converges to an element of the space
(C) Has a convergent subsequence only
(D) Converges only if it is monotone

Answer: (B)

Explanation: *By definition of a Banach space (complete NLS): every Cauchy sequence converges to an element of the space. This is both necessary and sufficient for completeness.*

Q19

Minkowski's inequality states: for $1 \leq p \leq \infty$,

(A) $\|xy\|_p \leq \|x\|_p \|y\|_p$
(B) $\|x + y\|_p \leq \|x\|_p + \|y\|_p$
(C) $\|x - y\|_p \geq |\|x\|_p - \|y\|_p|$
(D) $\|x\|_p \|y\|_p \leq \|x\|_2 \|y\|_2$

Answer: (B)

Explanation: *Minkowski's inequality is exactly the triangle inequality for the ℓ^p and L^p norms: $\|x+y\|_p \leq \|x\|_p + \|y\|_p$. This is what verifies that $\|\cdot\|_p$ is indeed a norm (the triangle inequality is the hardest axiom to check for ℓ^p).*

Q20

Which is the correct statement about ℓ^1 and ℓ^2?

(A) $\ell^2 \subseteq \ell^1$
(B) $\ell^1 \subseteq \ell^2$ and $\ell^1 \neq \ell^2$
(C) $\ell^1 = \ell^2$
(D) They are isometrically isomorphic

Answer: (B)

Explanation: *$\ell^1 \subsetneq \ell^2$: if $\sum |x_n| < \infty$ then $|x_n| \to 0$, so $|x_n| \leq 1$ eventually, hence $|x_n|^2 \leq |x_n|$ and $\sum |x_n|^2 < \infty$. The inclusion is strict: $(1/n) \in \ell^2 \setminus \ell^1$ since $\sum 1/n = \infty$ but $\sum 1/n^2 < \infty$.*

Q21

The space ℓ^∞ consists of:

(A) Absolutely summable sequences
(B) Square-summable sequences
(C) Bounded sequences with $\|(x_n)\|_\infty = \sup_n |x_n|$
(D) Sequences converging to zero

Answer: (C)

Explanation: *$\ell^\infty = \{(x_n) : \sup_n |x_n| < \infty\}$ with norm $\|(x_n)\|_\infty = \sup_n |x_n|$. (A) is ℓ^1, (B) is ℓ^2, (D) is $c_0 \subset \ell^\infty$.*

Q22

Which statement about finite-dimensional normed spaces is TRUE?

(A) They need not be complete
(B) All are Banach spaces
(C) Their unit ball is never compact
(D) They are all isomorphic to $\mathbb{R}$

Answer: (B)

Explanation: *Every finite-dimensional normed space is a Banach space (complete). Proof: all norms on a finite-dimensional space are equivalent to the Euclidean norm; $\mathbb{R}^n$ (Euclidean) is complete; hence so is any finite-dimensional NLS.*

Q23

The **completion** of $(C[0,1], \|\cdot\|_p)$ for $1 \leq p < \infty$ is:

(A) $(C[0,1], \|\cdot\|_\infty)$
(B) $L^p[0,1]$
(C) ℓ^p
(D) The space is already complete

Answer: (B)

Explanation: *The completion of $(C[0,1], \|\cdot\|_p)$ is $L^p[0,1]$, the space of p-integrable functions (Lebesgue). This is one motivation for defining L^p spaces: they complete the continuous functions under the L^p norm.*

Q24

If X is a Banach space and $Y \subseteq X$ is a **closed** subspace, then X/Y (the quotient space) with the quotient norm $\|[x]\| = \inf_{y\in Y} \|x - y\|$ is:

(A) A normed space but not necessarily Banach
(B) A Banach space
(C) A Hilbert space
(D) Not a normed space in general

Answer: (B)

Explanation: *The quotient of a Banach space by a closed subspace is again a Banach space. The quotient norm $\|[x]\| = \operatorname{dist}(x, Y)$ is indeed a norm (closedness of Y is essential for positive definiteness), and completeness of X passes to X/Y.*

Q25

A normed space X is **separable** if:

(A) It is finite-dimensional
(B) It contains a countable dense subset
(C) Every sequence has a convergent subsequence
(D) It is isomorphic to $\mathbb{R}^n$

Answer: (B)

Explanation: *Separable = has a countable dense subset. Examples: $\mathbb{R}^n$ (separable), $C[a,b]$ (polynomials with rational coefficients are dense by Weierstrass), ℓ^p for $1 \leq p < \infty$ (sequences with finitely many nonzero rational entries). ℓ^∞ is NOT separable.*

Q26

In which of the following does the unit ball fail to be compact?

(A) $\mathbb{R}^n$ with Euclidean norm
(B) $C[0,1]$ with sup norm
(C) $\mathbb{C}^2$ with $\|\cdot\|_2$
(D) $\mathbb{R}^3$

Answer: (B)

Explanation: *By Riesz's theorem: a normed space has compact unit ball $\iff$ it is finite-dimensional. $C[0,1]$ is infinite-dimensional, so its unit ball is NOT compact. The sequence $f_n(t) = \sin(n\pi t)$ lies in the unit ball but has no convergent subsequence.*

Q27

Which of the following norms on $\mathbb{R}^2$ are equivalent?

(A) Only $\|\cdot\|_1$ and $\|\cdot\|_2$
(B) Only $\|\cdot\|_2$ and $\|\cdot\|_\infty$
(C) All norms on $\mathbb{R}^2$ are equivalent
(D) No two distinct norms are equivalent

Answer: (C)

Explanation: *All norms on a finite-dimensional space are equivalent. On $\mathbb{R}^2$ specifically: $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{2}\|x\|_\infty$ and $\|x\|_2 \leq \|x\|_1 \leq \sqrt{2}\|x\|_2$, showing all three standard norms are pairwise equivalent (as are all other norms on $\mathbb{R}^2$).*

Q28

A **Schauder basis** for a Banach space X is a sequence (e_n) such that:

(A) Every bounded sequence is a multiple of some e_n

(B) Every $x \in X$ has a unique representation $x = \sum \alpha_n e_n$ (convergent in norm)

(C) (e_n) is an orthonormal set

(D) The e_n form a finite spanning set

Answer: (B)

Explanation: *Schauder basis: every element has a unique norm-convergent series expansion. Note: requires norm convergence, not just algebraic spanning. Standard example: $\{e_n\}$ (standard unit sequences) form a Schauder basis for ℓ^p, $1 \le p < \infty$.*

Q29

The dual space $(\ell^p)^*$ for $1 < p < \infty$ is isometrically isomorphic to:

(A) ℓ^p itself

(B) ℓ^q where $1/p + 1/q = 1$

(C) ℓ^∞

(D) ℓ^1

Answer: (B)

Explanation: *$(\ell^p)^* \cong \ell^q$ where q is the conjugate exponent ($1/p + 1/q = 1$). The duality pairing is $f(x) = \sum x_n y_n$ for $y \in \ell^q$. Special cases: $(\ell^1)^* \cong \ell^\infty$, $(\ell^2)^* \cong \ell^2$ (self-dual, Hilbert space).*

Q30

The space c_0 (sequences converging to 0, sup norm) has dual:

(A) c_0 itself

(B) ℓ^2

(C) ℓ^1

(D) ℓ^∞

Answer: (C)

Explanation: *$(c_0)^* \cong \ell^1$ and $(\ell^1)^* \cong \ell^\infty$. These are classical results. The identification is: each $f \in (c_0)^*$ corresponds to $y = (y_n) \in \ell^1$ via $f(x) = \sum x_n y_n$.*

Q31

Minkowski's inequality is used to prove which norm axiom for $\|\cdot\|_p$?

(A) Positive definiteness

(B) Absolute homogeneity

(C) Triangle inequality

(D) Symmetry

Answer: (C)

Explanation: *Minkowski's inequality $\|f+g\|_p \le \|f\|_p + \|g\|_p$ is exactly the triangle inequality. It is derived from Hölder's inequality. The other axioms (positive definiteness, homogeneity) are straightforward from the definition.*

Q32

The space $L^\infty[0,1]$ is:

(A) Not a normed space

(B) A normed space but not Banach

(C) A Banach space

(D) A Hilbert space

Answer: (C)

Explanation: *$L^\infty[0,1]$ (essentially bounded measurable functions with norm $\|f\|_\infty = \operatorname{ess\,sup}|f|$) is a Banach space. It is NOT a Hilbert space (the norm does not come from an inner product; the parallelogram law fails).*

Q33

An **isometric isomorphism** between normed spaces X and Y is a:

(A) Bijective linear map preserving norms: $\|Tx\| = \|x\|$ for all x

(B) Continuous bijection

(C) Linear map with bounded inverse

(D) Surjective isometry

Answer: (A)

Explanation: *An isometric isomorphism is a bijective linear map $T : X \to Y$ that preserves norms: $\|Tx\|_Y = \|x\|_X$. Such a map is automatically injective (if $Tx = 0$ then $\|x\| = 0$ so $x = 0$). Isometric isomorphisms are the "equality" relation between normed spaces.*

Q34

The **parallelogram law** in a normed space states:

(A) $\|x+y\|^2+\|x-y\|^2 = 2(\|x\|^2+\|y\|^2)$
(B) $\|x+y\|^2 = \|x\|^2+\|y\|^2$
(C) $\|x+y\| \le \|x\|+\|y\|$
(D) $\|\alpha x\| = |\alpha|\|x\|$

Answer: (A)

Explanation: *The parallelogram law: $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2+\|y\|^2)$. A normed space satisfies the parallelogram law $\iff$ the norm comes from an inner product (Jordan-von Neumann theorem). This characterizes Hilbert spaces among Banach spaces.*

Q35

If $\|x_n - x\| \to 0$ and $\|x_n - y\| \to 0$, then:

(A) $x = y$ always
(B) $x = y$ only in Hausdorff spaces
(C) x may differ from y
(D) $x - y$ is bounded

Answer: (A)

Explanation: *Limits are unique in normed spaces. Proof: $\|x-y\| \le \|x-x_n\|+\|x_n-y\| \to 0$, so $\|x-y\| = 0$, hence $x = y$. (This reflects that every normed space is Hausdorff as a metric space.)*

Q36

The **Hahn-Banach theorem** guarantees the existence of:

(A) A complete norm on every vector space
(B) Nontrivial bounded linear functionals on any normed space
(C) An inner product on every Banach space
(D) A Schauder basis for every Banach space

Answer: (B)

Explanation: *The Hahn-Banach theorem (extension form) guarantees that nontrivial bounded linear functionals exist on any normed space and can be extended from subspaces. This separates points: for each $x \neq 0$, there exists $f \in X^*$ with $f(x) \neq 0$.*

Q37

The **reverse triangle inequality** in a normed space is:

(A) $\|x+y\| \ge \|x\| - \|y\|$
(B) $\big|\|x\| - \|y\|\big| \le \|x-y\|$
(C) $\|x-y\| \ge \|x\|+\|y\|$
(D) $\|x\| \le \|x-y\|+\|y\|$

Answer: (B)

Explanation: *Reverse triangle inequality: $\big|\|x\| - \|y\|\big| \le \|x-y\|$. Proof: $\|x\| = \|(x-y)+y\| \le \|x-y\|+\|y\|$, so $\|x\|-\|y\| \le \|x-y\|$; symmetrically $\|y\|-\|x\| \le \|x-y\|$. (D) is just the triangle inequality rearranged.*

Q38

Which normed space is **reflexive** (isometrically isomorphic to its bidual X^{**})?

(A) ℓ^1
(B) ℓ^∞
(C) ℓ^2
(D) c_0

Answer: (C)

Explanation: *ℓ^2 is a Hilbert space, hence reflexive ($\ell^2 \cong (\ell^2)^* \cong (\ell^2)^{**}$). Also ℓ^p is reflexive for $1 < p < \infty$. ℓ^1 is not reflexive ($(\ell^1)^{**} = (\ell^\infty)^* \supsetneq \ell^1$). ℓ^∞ and c_0 are also not reflexive.*

Q39

The **open unit ball** in a normed space is:

(A) $\{x : \|x\| \le 1\}$
(B) $\{x : \|x\| < 1\}$
(C) $\{x : \|x\| = 1\}$
(D) $\{x : x \neq 0\}$

Answer: (B)

Explanation: *Open unit ball: $B(0,1) = \{x \in X : \|x\| < 1\}$ (strict inequality). Closed unit ball: $\bar{B}(0,1) = \{x : \|x\| \le 1\}$. Unit sphere: $S = \{x : \|x\| = 1\}$. In Riesz's theorem, it is the closed unit ball whose compactness characterizes finite-dimensionality.*

Q40

If X is a Banach space and $\dim X = \infty$, then:

(A) The closed unit ball is compact
(B) All bounded sequences have convergent subsequences

(C) The closed unit ball is NOT compact

(D) Every bounded linear operator is compact

Answer: (C)

Explanation: *By Riesz's theorem: infinite-dimensional normed space $\Rightarrow$ closed unit ball is NOT compact. Riesz's lemma shows: for any proper closed subspace $Y \subsetneq X$, there exists x with $\|x\| = 1$ and $\mathrm{dist}(x, Y) > 1/2$, giving a sequence with no convergent subsequence.*

[DAY] DAY 6 — Inner Product & Hilbert Spaces

Topics: Inner product spaces, Cauchy-Schwarz inequality, Hilbert spaces, orthogonality, orthonormal sets, Gram-Schmidt, Bessel's inequality, Parseval's theorem.

2. Inner Product Spaces

[DEF] Inner Product Space

A **inner product** on a vector space X over $\mathbb{K}$ is a map $\langle\cdot,\,\cdot\rangle : X \times X \to \mathbb{K}$ satisfying:

(IP1) $\langle x,\, x\rangle \geq 0$ and $\langle x,\, x\rangle = 0 \iff x = 0$

(IP2) $\langle \alpha x + \beta y,\, z\rangle = \alpha\langle x,\, z\rangle + \beta\langle y,\, z\rangle$ (linearity in first slot)

(IP3) $\langle x,\, y\rangle = \overline{\langle y,\, x\rangle}$ (conjugate symmetry)

The pair $(X, \langle\cdot,\,\cdot\rangle)$ is an **inner product space**. Every inner product induces a norm: $\|x\| = \sqrt{\langle x,\, x\rangle}$.

[THM] Cauchy-Schwarz Inequality

In an inner product space $(X, \langle\cdot,\,\cdot\rangle)$:

$$|\langle x,\, y\rangle| \;\leq\; \|x\| \cdot \|y\| \quad \forall\, x, y \in X.$$

Equality holds $\iff$ x and y are linearly dependent ($x = \alpha y$ for some scalar α).

[PROOF] Proof of Cauchy-Schwarz Inequality

If $y = 0$: both sides are 0. Suppose $y \neq 0$. For any $\lambda \in \mathbb{K}$: $0 \leq \langle x - \lambda y,\, x - \lambda y\rangle = \|x\|^2 - \lambda\overline{\langle x,\, y\rangle} - \bar{\lambda}\langle x,\, y\rangle + |\lambda|^2\|y\|^2$. Set $\lambda = \langle x,\, y\rangle/\|y\|^2$: $0 \leq \|x\|^2 - \frac{|\langle x,y\rangle|^2}{\|y\|^2}$. Hence $|\langle x,\, y\rangle|^2 \leq \|x\|^2\|y\|^2$, giving $|\langle x,\, y\rangle| \leq \|x\|\|y\|$. □

[DEF] Hilbert Space

A **Hilbert space** is an inner product space that is **complete** with respect to the norm $\|x\| = \sqrt{\langle x,\, x\rangle}$.

[EX] Standard Hilbert Spaces

- $\mathbb{R}^n$ with $\langle x,\, y\rangle = \sum x_i y_i$.
- ℓ^2 with $\langle x,\, y\rangle = \sum x_n \bar{y}_n$.
- $L^2[a, b]$ with $\langle f,\, g\rangle = \int_a^b f(t)\overline{g(t)}\, dt$.
- Any finite-dimensional inner product space.

NOT Hilbert: ℓ^p for $p \neq 2$; L^p for $p \neq 2$; $C[a, b]$ with any standard norm.

[THM] Jordan-von Neumann Theorem

A Banach space $(X, \|\cdot\|)$ is a Hilbert space (i.e., the norm comes from an inner product) $\iff$ the **parallelogram law** holds:

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2) \quad \forall\, x, y \in X.$$

When it holds, the inner product is recovered by the **polarization identity**:

$$\langle x, y\rangle = \tfrac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2\right) \quad \text{(real case)}.$$

2.1. Orthogonality and Orthonormal Sets

[DEF] Orthogonality

In a Hilbert space H:

- $x \perp y$ (x is orthogonal to y) if $\langle x, y\rangle = 0$.
- $A \perp B$ if $\langle a, b\rangle = 0$ for all $a \in A$, $b \in B$.
- The **orthogonal complement** of A: $A^\perp = \{x \in H : \langle x, a\rangle = 0\ \forall\, a \in A\}$.
- A set $\{e_\alpha\}$ is **orthonormal** if $\langle e_\alpha, e_\beta\rangle = \delta_{\alpha\beta}$ (Kronecker delta).

[THM] Orthogonal Decomposition Theorem

Let M be a **closed** subspace of a Hilbert space H. Then:

$H = M \oplus M^\perp$ (every $x \in H$ has unique decomposition $x = m + m^\perp$, $m \in M$, $m^\perp \in M^\perp$).

The map $P : H \to M$ sending $x \mapsto m$ is the **orthogonal projection** onto M: $P^2 = P$, $P^* = P$ (self-adjoint), $\|Px\| \le \|x\|$ (bounded).

[THM] Gram-Schmidt Process

Given a linearly independent sequence $\{v_1, v_2, v_3, \ldots\}$ in an inner product space, define:

$$e_1 = \frac{v_1}{\|v_1\|}, \qquad u_n = v_n - \sum_{k=1}^{n-1} \langle v_n, e_k\rangle e_k, \qquad e_n = \frac{u_n}{\|u_n\|}.$$

Then $\{e_n\}$ is an orthonormal sequence with $\operatorname{span}\{e_1, \ldots, e_n\} = \operatorname{span}\{v_1, \ldots, v_n\}$ for each n.

[THM] Bessel's Inequality and Parseval's Theorem

Let $\{e_n\}$ be an orthonormal sequence in a Hilbert space H.

Bessel's inequality: $\displaystyle\sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2 \le \|x\|^2 \quad \forall\, x \in H.$

Parseval's theorem (when $\{e_n\}$ is a **complete** orthonormal system): $\displaystyle\sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2 = \|x\|^2 \quad \forall\, x \in H.$

The scalars $\hat{x}_n = \langle x, e_n\rangle$ are the **Fourier coefficients** of x.

[DEF] Complete Orthonormal System (CONS)

An orthonormal set $\{e_\alpha\}$ in a Hilbert space H is **complete** (or a **Hilbert basis**) if: $\langle x, e_\alpha\rangle = 0$ for all α implies $x = 0$. Equivalently, $\operatorname{span}\{e_\alpha\}$ is dense in H.

Day 6 MCQs — Inner Product & Hilbert Spaces

Q41

An inner product on X must satisfy:
(A) Linearity in both arguments, positivity, symmetry
(B) Positive definiteness, linearity in first argument, conjugate symmetry
(C) Bilinearity (linear in both) and positivity
(D) Non-negativity, commutativity, and homogeneity

Answer: (B)

Explanation: *Inner product: (IP1)* $\langle x, y\rangle$...

Q42

The **Cauchy-Schwarz inequality** states:
(A) $\|x + y\| \le \|x\| + \|y\|$
(B) $|\langle x, y\rangle| \le \|x\| \cdot \|y\|$
(C) $\|x\|^2 + \|y\|^2 \ge 2|\langle x, y\rangle|$
(D) $\langle x, y\rangle^2 \le \|x\|^2 + \|y\|^2$

Answer: (B)

Explanation: *Cauchy-Schwarz:* $|\langle x, y\rangle| \le \|x\|\|y\|$. *Equality holds iff* x *and* y *are linearly dependent. This is the most important inequality in Hilbert space theory. It is used to prove the triangle inequality for the induced norm.*

Q43

A **Hilbert space** is an inner product space that is:
(A) Finite-dimensional
(B) Separable
(C) Complete with respect to the induced norm
(D) Locally compact

Answer: (C)

Explanation: *Hilbert space = inner product space + complete (with respect to the norm induced by the inner product). Every finite-dimensional inner product space is Hilbert;* $L^2[a, b]$ *and* ℓ^2 *are infinite-dimensional Hilbert spaces.*

Q44

Which of the following is a Hilbert space?
(A) $(C[0, 1], \|\cdot\|_\infty)$
(B) $(\ell^1, \|\cdot\|_1)$
(C) $(L^2[0, 1], \langle\cdot, \cdot\rangle)$
(D) $(\ell^\infty, \|\cdot\|_\infty)$

Answer: (C)

Explanation: $L^2[0, 1]$ *with* $\langle f, g\rangle = \int_0^1 f\bar{g}$ *is a Hilbert space (complete inner product space).* $C[0, 1]$ *with* $\|\cdot\|_\infty$ *satisfies the triangle inequality but NOT the parallelogram law.* ℓ^1 *and* ℓ^∞ *are Banach but not Hilbert.*

Q45

The parallelogram law is:

(A) $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$
(B) $\|x+y\|^2 = \|x\|^2 + \|y\|^2$
(C) $\|x+y\|^2 - \|x-y\|^2 = 4\langle x, y\rangle$
(D) $\|x\|^2 + \|y\|^2 = \langle x, y\rangle^2$

Answer: (A)

Explanation: *Parallelogram law:* $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$. *By Jordan-von Neumann: a Banach space is Hilbert* $\iff$ *this law holds. Note (C) is the polarization identity (over* $\mathbb{R}$*):* $\langle x, y\rangle = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2)$.

Q46

$x \perp y$ in a Hilbert space means:

(A) $\|x-y\| = 0$
(B) $\langle x, y\rangle = 0$
(C) $\|x+y\| = \|x\| + \|y\|$
(D) $x + y = 0$

Answer: (B)

Explanation: *Orthogonality:* $x \perp y \iff \langle x, y\rangle = 0$. *Equivalent to the Pythagorean theorem:* $x \perp y \iff \|x+y\|^2 = \|x\|^2 + \|y\|^2$. *(C) is the condition for equality in the triangle inequality, equivalent to* $x = \lambda y$ *for some* $\lambda \geq 0$.

Q47

The **Pythagorean theorem** in a Hilbert space: if $x \perp y$, then:

(A) $\|x+y\| = \|x\| + \|y\|$
(B) $\|x+y\|^2 = \|x\|^2 + \|y\|^2$
(C) $\|x+y\|^2 = \|x\|^2 - \|y\|^2$
(D) $\langle x+y, x+y\rangle = \langle x, y\rangle$

Answer: (B)

Explanation: *Pythagoras in Hilbert space:* $x \perp y \Rightarrow \|x+y\|^2 = \|x\|^2 + \|y\|^2$. *Proof:* $\|x+y\|^2 = \langle x+y, x+y\rangle = \|x\|^2 + \langle x, y\rangle + \langle y, x\rangle + \|y\|^2 = \|x\|^2 + 0 + 0 + \|y\|^2$.

Q48

An **orthonormal set** $\{e_n\}$ satisfies:

(A) $\langle e_m, e_n\rangle = 1$ for all m, n
(B) $\langle e_m, e_n\rangle = \delta_{mn}$ (Kronecker delta)
(C) $e_m + e_n = 0$ for $m \neq n$
(D) $\|e_m + e_n\| = 2$ for all m, n

Answer: (B)

Explanation: *Orthonormal:* $\langle e_m, e_n\rangle = \delta_{mn} = \begin{cases} 1 & m = n \\ 0 & m \neq n \end{cases}$. *Each* e_n *has unit norm and any two distinct elements are orthogonal. Standard example:* $e_n = (0, \ldots, 1, \ldots, 0)$ *(1 in position* n*) in* ℓ^2.

Q49

Bessel's inequality states: for an orthonormal sequence $\{e_n\}$ in Hilbert H:

(A) $\sum |\langle x, e_n\rangle|^2 = \|x\|^2$
(B) $\sum |\langle x, e_n\rangle|^2 \leq \|x\|^2$
(C) $\sum |\langle x, e_n\rangle| \leq \|x\|$
(D) $\sup |\langle x, e_n\rangle| \leq \|x\|$

Answer: (B)

Explanation: *Bessel's inequality:* $\sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2 \leq \|x\|^2$. *Equality (Parseval) holds iff* $\{e_n\}$ *is a complete orthonormal system (Hilbert basis). The inequality guarantees the series of Fourier coefficients is always convergent.*

Q50

Parseval's theorem (for a complete ONS $\{e_n\}$) states:

(A) $\sum |\langle x, e_n\rangle|^2 \leq \|x\|^2$
(B) $\sum |\langle x, e_n\rangle|^2 = \|x\|^2$ for all $x \in H$
(C) $x = \sum \langle x, e_n\rangle e_n$ in the algebraic sense only
(D) $\|x\| = \sum |\langle x, e_n\rangle|$

Answer: (B)

Explanation: *Parseval's theorem: when* $\{e_n\}$ *is a complete (total) ONS,* $\sum |\langle x, e_n\rangle|^2 = \|x\|^2$. *This is equality in Bessel's inequality. Also:* $x = \sum \langle x, e_n\rangle e_n$ *(convergent in* H*). This generalizes Fourier series:* $\|f\|_{L^2}^2 = \sum |\hat{f}_n|^2$.

Q51

The Gram-Schmidt process takes a linearly independent set and produces:

(A) A basis for the whole Hilbert space
(B) An orthonormal set with the same span
(C) The orthogonal complement
(D) A minimal Cauchy sequence

Answer: (B)

Explanation: *Gram-Schmidt: starting from* $\{v_1, v_2, \ldots\}$ *linearly independent, produces orthonormal* $\{e_1, e_2, \ldots\}$ *with* $\mathrm{span}\{e_1, \ldots, e_n\} = \mathrm{span}\{v_1, \ldots, v_n\}$ *for each* n. *The inner product structure is used to "subtract off" previously computed directions.*

Q52

For a closed subspace M of Hilbert H, the **orthogonal complement** $M^\perp$ satisfies:

(A) $M \cap M^\perp = M$
(B) $H = M \oplus M^\perp$ (every element decomposes uniquely)
(C) $M^\perp = H$
(D) $M \cup M^\perp = H$

Answer: (B)

Explanation: *Orthogonal decomposition:* $H = M \oplus M^\perp$, *i.e., every* $x \in H$ *decomposes uniquely as* $x = m + m^\perp$ *with* $m \in M$, $m^\perp \in M^\perp$, *and* $m \perp m^\perp$. *This requires* M *to be closed.* $M \cap M^\perp = \{0\}$.

Q53

The **orthogonal projection** $P : H \to M$ satisfies:

(A) $P^2 = P$, $P^* = P$, $\|P\| = 1$ (if $M \neq \{0\}$)
(B) $P^2 = 0$
(C) P is surjective and injective
(D) $\|Px\| = \|x\|$ for all x

Answer: (A)

Explanation: *Orthogonal projection onto closed $M \subseteq H$: $P^2 = P$ (idempotent), $P^* = P$ (self-adjoint), $\ker P = M^\perp$, $\operatorname{ran} P = M$, and $\|P\| = 1$ (if $M \neq \{0\}$). Also, Px is the closest point in M to x.*

Q54

In $L^2[0, 2\pi]$, the functions $\{e^{int}/\sqrt{2\pi} : n \in \mathbb{Z}\}$ form:

(A) An orthonormal basis (CONS)
(B) A Schauder basis only
(C) A linearly dependent set
(D) A closed subspace

Answer: (A)

Explanation: *$\{e^{int}/\sqrt{2\pi} : n \in \mathbb{Z}\}$ is a complete orthonormal system (Hilbert basis) for $L^2[0, 2\pi]$. This is the basis for Fourier series: every $f \in L^2[0, 2\pi]$ can be expanded as $f = \sum \hat{f}_n e^{int}/\sqrt{2\pi}$ with convergence in L^2.*

Q55

Every Hilbert space is:

(A) Separable
(B) A Banach space
(C) Finite-dimensional
(D) Isometrically isomorphic to ℓ^2

Answer: (B)

Explanation: *Every Hilbert space is a Banach space (complete normed space, where the norm comes from an inner product). Not every Hilbert space is separable (there exist non-separable Hilbert spaces). Not every Banach space is Hilbert (e.g., ℓ^1).*

Q56

The dual space of a Hilbert space H is isometrically isomorphic to:

(A) H itself (Riesz representation)
(B) ℓ^2
(C) L^∞
(D) The orthogonal complement

Answer: (A)

Explanation: *Riesz Representation Theorem: for every bounded linear functional $f : H \to \mathbb{K}$ on a Hilbert space, $\exists! y \in H$ with $f(x) = \langle x, y \rangle$ and $\|f\| = \|y\|$. So $H^* \cong H$ isometrically. This is why Hilbert spaces are reflexive and "self-dual."*

Q57

If $\{e_n\}$ is an ONS (not necessarily complete) in Hilbert H, then for all $x \in H$:

(A) $x = \sum\langle x, e_n\rangle e_n$
(B) $\sum|\langle x, e_n\rangle|^2 \le \|x\|^2$ (Bessel)
(C) $\langle x, e_n\rangle = 0$ for all n
(D) $\|x\| = \sum\langle x, e_n\rangle$

Answer: (B)

Explanation: *Without completeness, only Bessel's inequality holds: $\sum|\langle x, e_n\rangle|^2 \le \|x\|^2$. The expansion $x = \sum\langle x, e_n\rangle e_n$ (in H) requires the ONS to be complete (a Hilbert basis).*

Q58

In an inner product space, $\|x+y\|^2 = \|x\|^2 + \|y\|^2$ implies:

(A) $x = y$
(B) $x \perp y$
(C) $\langle x, y\rangle = \|x\|\|y\|$
(D) $x = 0$ or $y = 0$

Answer: (B)

Explanation: *$\|x+y\|^2 = \|x\|^2 + 2\Re\langle x, y\rangle + \|y\|^2 = \|x\|^2 + \|y\|^2 \iff \Re\langle x, y\rangle = 0$. Over $\mathbb{R}$, this means $\langle x, y\rangle = 0$, i.e., $x \perp y$. This is the Pythagorean theorem; the converse also holds: $x \perp y \Rightarrow \|x+y\|^2 = \|x\|^2 + \|y\|^2$.*

Q59

The best approximation in a closed subspace $M \subseteq H$ to a point $x \in H$ is:

(A) Any point of M
(B) The orthogonal projection Px
(C) The point x itself
(D) The midpoint of x and the origin

Answer: (B)

Explanation: *The closest point in M to x is the orthogonal projection Px. Proof: for any $m \in M$, $\|x-m\|^2 = \|x-Px\|^2 + \|Px-m\|^2 \ge \|x-Px\|^2$. So Px minimizes $\|x-m\|$ over $m \in M$.*

Q60

A separable Hilbert space is isometrically isomorphic to:

(A) $\mathbb{R}^n$ for some n
(B) ℓ^2
(C) $L^\infty[0,1]$
(D) $C[0,1]$

Answer: (B)

Explanation: *Every separable infinite-dimensional Hilbert space is isometrically isomorphic to ℓ^2. This is the "universal" separable Hilbert space. Finite-dimensional Hilbert spaces are isomorphic to $\mathbb{R}^n$ (or $\mathbb{C}^n$) for some n.*

Q61

In ℓ^2, the standard basis $\{e_n\}$ (with 1 in position n, 0 elsewhere) is:
(A) A complete orthonormal system
(B) An orthogonal but not orthonormal set
(C) Linearly dependent
(D) Dense in ℓ^1 but not ℓ^2

Answer: (A)

Explanation: *$\{e_n\}$ in ℓ^2: $\langle e_m, e_n\rangle = \delta_{mn}$ (orthonormal). For any $x = (x_n) \in \ell^2$: $\langle x, e_n\rangle = x_n$, so Parseval: $\sum |x_n|^2 = \|x\|^2$. And $x = \sum x_n e_n$ in ℓ^2. So $\{e_n\}$ is a CONS.*

Q62

$(M^\perp)^\perp$ for a closed subspace $M \subseteq H$ equals:
(A) H
(B) $M^\perp$
(C) M
(D) $\{0\}$

Answer: (C)

Explanation: *$(M^\perp)^\perp = M$ when M is closed. Since $H = M \oplus M^\perp$, taking orthogonal complement of both sides and using $(M^\perp)^\perp = M$. If M is not closed, $(M^\perp)^\perp = \overline{M}$.*

Q63

Which of the following satisfies the parallelogram law?
(A) $(\ell^1, \|\cdot\|_1)$
(B) $(C[0,1], \|\cdot\|_\infty)$
(C) $(\ell^2, \|\cdot\|_2)$
(D) $(\mathbb{R}^n, \|\cdot\|_1)$ for $n \geq 2$

Answer: (C)

Explanation: *ℓ^2 is a Hilbert space, so it satisfies the parallelogram law. ℓ^1 does not (take $x = (1,0,0,\ldots)$, $y = (0,1,0,\ldots)$: $\|x+y\|_1^2 + \|x-y\|_1^2 = 4 \neq 2(\|x\|_1^2 + \|y\|_1^2) = 4$... actually equals 4 here, but fails for other vectors). The Jordan-von Neumann theorem confirms ℓ^p is Hilbert iff $p = 2$.*

Q64

The Riesz representation theorem for Hilbert spaces states:
(A) Every bounded operator on H has an adjoint
(B) Every bounded functional $f : H \to \mathbb{K}$ is of the form $f(x) = \langle x, y\rangle$ for unique $y \in H$
(C) Every Hilbert space has a Schauder basis
(D) $H^{**} \cong H$

Answer: (B)

Explanation: *Riesz Representation Theorem: $H^* \cong H$ via $y \mapsto f_y$ where $f_y(x) = \langle x, y\rangle$. The map $y \mapsto f_y$ is a conjugate-linear isometric isomorphism $H \to H^*$. This makes Hilbert spaces self-dual and reflexive.*

Q65

In the Gram-Schmidt formula, $u_n = v_n - \sum_{k=1}^{n-1} \langle v_n, e_k\rangle e_k$ represents:
(A) The projection of v_n onto e_{n-1}

(B) v_n with its components along $e_1, \dots, e_{n-1}$ removed
(C) The orthogonal complement of v_n
(D) The normalization of v_n

Answer: (B)

Explanation: *In Gram-Schmidt, $\langle v_n, e_k \rangle e_k$ is the component of v_n in the direction e_k. Subtracting all such components leaves u_n, which is orthogonal to all $e_1, \dots, e_{n-1}$. Then $e_n = u_n/\|u_n\|$ normalizes it.*

Q66

If $f : H \to \mathbb{R}$ is a bounded linear functional and $\ker f \neq H$, then $\ker f$:
(A) Is a dense subspace of H
(B) Is a closed hyperplane (codimension 1 closed subspace)
(C) Is a compact subspace
(D) Equals $\{0\}$

Answer: (B)

Explanation: *The kernel of a bounded (continuous) linear functional on H is closed (preimage of closed set $\{0\}$ under continuous map) and has codimension 1 (a hyperplane). If $f \neq 0$, $\ker f$ is a proper closed subspace of codimension 1.*

Q67

For orthonormal $\{e_1, e_2, \dots, e_N\}$ in H, the best N-term approximation to x is:
(A) $\frac{1}{N} \sum_{n=1}^{N} e_n$
(B) $\sum_{n=1}^{N} \langle x, e_n \rangle e_n$
(C) $\sum_{n=1}^{N} e_n$
(D) $N e_N$

Answer: (B)

Explanation: *The best approximation of x from $\operatorname{span}\{e_1, \dots, e_N\}$ is the orthogonal projection $P_N x = \sum_{n=1}^{N} \langle x, e_n \rangle e_n$. The coefficients $\langle x, e_n \rangle$ are the Fourier coefficients. This minimizes $\|x - \sum c_n e_n\|$ over all choices of scalars c_n.*

Q68

An inner product space is a Hilbert space if and only if:
(A) It is finite-dimensional
(B) It satisfies the parallelogram law
(C) It is complete in the norm induced by the inner product
(D) It has a countable orthonormal basis

Answer: (C)

Explanation: *By definition: Hilbert space = inner product space + complete (in the induced norm). (A) is sufficient but not necessary. (B) characterizes when a Banach norm comes from an inner product. (D) characterizes separable Hilbert spaces.*

Q69

The Fourier coefficients $\hat{x}_n = \langle x, e_n \rangle$ in Parseval's theorem satisfy:
(A) $(\hat{x}_n) \in \ell^\infty$ always
(B) $(\hat{x}_n) \in \ell^2$ and $\sum |\hat{x}_n|^2 = \|x\|^2$
(C) $\hat{x}_n = \langle e_n, x \rangle$

(D) $\sum \hat{x}_n = \|x\|$

Answer: (B)

Explanation: *Parseval: the sequence of Fourier coefficients $(\hat{x}_n) = (\langle x, e_n \rangle)$ belongs to ℓ^2 and $\sum |\hat{x}_n|^2 = \|x\|^2$. This is an isometry: the map $x \mapsto (\hat{x}_n)$ is an isometric isomorphism from a separable Hilbert space to ℓ^2.*

Q70

Which of the following spaces is NOT a Hilbert space?

(A) $(\ell^2, \langle \cdot, \cdot \rangle)$

(B) $(L^2[0,1], \langle \cdot, \cdot \rangle)$

(C) $(C[0,1], \|\cdot\|_\infty)$

(D) $(\mathbb{R}^5, \text{Euclidean})$

Answer: (C)

Explanation: *$(C[0,1], \|\cdot\|_\infty)$ is a Banach space but NOT a Hilbert space: the parallelogram law fails. Take $f = 1$ (constant 1) and $g = t$: $\|f+g\|_\infty^2 + \|f-g\|_\infty^2 = 4 + 1 = 5 \neq 2(1+1) = 4$.*

[DAY] DAY 7 — Hahn-Banach Theorem ★ (Very Important)

Topics: Bounded linear functionals, dual spaces, Hahn-Banach extension, geometric form (separation of convex sets), applications.

3. Bounded Linear Functionals & Dual Spaces

[DEF] Linear Functional and Bounded Linear Functional

A **linear functional** on X is a linear map $f : X \to \mathbb{K}$. f is **bounded** (equivalently, **continuous**) if:

$$\|f\| = \sup_{\|x\|\le 1} |f(x)| < \infty.$$

The **dual space** of X is $X^* = \{f : X \to \mathbb{K} : f \text{ bounded linear}\}$ with norm $\|f\|_{X^*} = \sup_{\|x\|\le 1} |f(x)|$. $(X^*, \|\cdot\|_{X^*})$ is always a Banach space.

[THM] Hahn-Banach Extension Theorem ★

Let X be a normed space, $Y \subseteq X$ a subspace, and $f : Y \to \mathbb{K}$ a bounded linear functional. Then there exists a bounded linear functional $F : X \to \mathbb{K}$ such that:

$$F\big|_Y = f \quad \text{(extension)} \qquad \text{and} \qquad \|F\|_{X^*} = \|f\|_{Y^*}.$$

(The extension preserves the norm.)

[THM] Geometric Hahn-Banach (Separation Theorem)

Let A and B be non-empty disjoint convex subsets of a normed space X.

(i) If A is open, there exists $f \in X^*$ and $\gamma \in \mathbb{R}$ such that $f(x) < \gamma \le f(y)$ for all $x \in A$, $y \in B$.

(ii) If A is compact and B is closed, there exists $f \in X^*$ and $\gamma_1 < \gamma_2$ such that $f(x) < \gamma_1 < \gamma_2 < f(y)$ for all $x \in A$, $y \in B$.

(A hyperplane "separates" the two convex sets.)

[COR] Key Consequences of Hahn-Banach

(i) For every $x \neq 0$ in X, $\exists\, f \in X^*$ with $f(x) = \|x\|$ and $\|f\| = 1$. (Dual space separates points.)

(ii) $\|x\| = \sup\{|f(x)| : f \in X^*, \|f\| \le 1\}$ (norm recovered from dual).

(iii) Every closed subspace is the intersection of kernels of bounded functionals.

(iv) Every normed space embeds isometrically into its bidual X^{**}.

(v) Closed convex sets are weakly closed (separation from exterior points).

Day 7 MCQs — Hahn-Banach Theorem

Q71

A linear functional $f : X \to \mathbb{K}$ is **bounded** if:

(A) $f(x) \geq 0$ for all x

(B) $\sup_{\|x\| \leq 1} |f(x)| < \infty$

(C) f is injective

(D) $\ker f = \{0\}$

Answer: (B)

Explanation: *Bounded linear functional:* $\|f\| = \sup_{\|x\| \leq 1} |f(x)| < \infty$. *Equivalently:* f *is continuous; or* $\exists C > 0$ *s.t.* $|f(x)| \leq C\|x\|$ *for all* x. *The smallest such* C *is* $\|f\|$.

Q72

A linear functional $f : X \to \mathbb{K}$ is continuous $\iff$:

(A) f is bijective

(B) f is bounded

(C) $\ker f = X$

(D) f is an isometry

Answer: (B)

Explanation: *For linear functionals: continuous* $\iff$ *bounded. This equivalence holds for all linear maps between normed spaces: continuous at 0* $\iff$ *bounded* $\iff$ *continuous everywhere. So continuity of linear maps is purely algebraic.*

Q73

The **dual space** X^* is always:

(A) Isomorphic to X

(B) A Banach space

(C) Finite-dimensional

(D) A Hilbert space

Answer: (B)

Explanation: X^* *is always a Banach space, even if* X *itself is not complete. Proof:* X^* *is isometrically embedded in the space of all bounded functions* $X \to \mathbb{K}$, *which is complete;* X^* *is a closed subspace, hence complete.*

Q74

The **Hahn-Banach theorem** guarantees:

(A) Every functional on a subspace can be extended to the whole space, preserving the norm

(B) Every Banach space has a Schauder basis

(C) Every bounded operator has an adjoint

(D) The dual of ℓ^p is ℓ^q

Answer: (A)

Explanation: *Hahn-Banach (extension form): bounded linear* $f : Y \to \mathbb{K}$ *on subspace* $Y \subseteq X$ *extends to* $F : X \to \mathbb{K}$ *with* $\|F\| = \|f\|$. *This is one of the three most important theorems in functional analysis (along with the Big Three).*

Q75

By Hahn-Banach, for any $x_0 \neq 0$ in normed X, $\exists\, f \in X^*$ with:

(A) $f(x_0) = 0$ and $\|f\| = 1$
(B) $f(x_0) = \|x_0\|$ and $\|f\| = 1$
(C) $f(x_0) = 1$ and $\|f\| = \|x_0\|$
(D) $f = 0$

Answer: (B)

Explanation: *HB corollary: $\forall\, x_0 \neq 0$, $\exists\, f \in X^*$ with $f(x_0) = \|x_0\|$ and $\|f\| = 1$. Proof: on span$\{x_0\}$, define $f_0(\alpha x_0) = \alpha\|x_0\|$; then $\|f_0\| = 1$; extend by HB. This shows $\|x\| = \sup_{\|f\|\leq 1} |f(x)|$.*

Q76

The **geometric** Hahn-Banach theorem is about:

(A) Extending linear functionals
(B) Separating disjoint convex sets by a hyperplane
(C) The completeness of the dual space
(D) The reflexivity of Hilbert spaces

Answer: (B)

Explanation: *Geometric HB (Separation theorem): two disjoint convex sets (one open, or one compact and one closed) can be separated by a closed hyperplane $\{x : f(x) = \gamma\}$. This is fundamental in optimization (duality theory) and convex analysis.*

Q77

The **bidual** $X^{**} = (X^*)^*$ always contains:

(A) Only the zero functional
(B) An isometric copy of X (via the canonical embedding $J : X \to X^{**}$)
(C) An isomorphic but not isometric copy of X
(D) X if and only if X is finite-dimensional

Answer: (B)

Explanation: *The canonical embedding $J : X \to X^{**}$, $Jx(f) = f(x)$, is an isometric linear injection. So X is always isometrically embedded in X^{**}. If J is surjective ($J(X) = X^{**}$), X is called **reflexive**.*

Q78

X is **reflexive** if:

(A) $X \cong X^*$
(B) The canonical embedding $J : X \to X^{**}$ is surjective
(C) X^* is separable
(D) $X = X^{**}$ as sets

Answer: (B)

Explanation: *Reflexive: the canonical map $J : X \to X^{**}$ is surjective (hence an isometric isomorphism). Reflexive spaces: all Hilbert spaces, ℓ^p and L^p for $1 < p < \infty$. Non-reflexive: ℓ^1, ℓ^∞, c_0, $C[0,1]$, L^1, L^∞.*

Q79

The norm of a bounded linear functional f on X is:

(A) $\|f\| = \inf_{\|x\|\leq 1} |f(x)|$

(B) $\|f\| = \sup_{\|x\|\le 1} |f(x)| = \sup_{\|x\|=1} |f(x)|$
(C) $\|f\| = |f(0)|$
(D) $\|f\| = \sup_{x\in X} |f(x)|$

Answer: (B)

Explanation: *$\|f\| = \sup_{\|x\|\le 1} |f(x)| = \sup_{\|x\|=1} |f(x)| = \sup_{x\neq 0} \frac{|f(x)|}{\|x\|}$. All three expressions are equal. (D) would be $+\infty$ for any nonzero functional.*

Q80

By Hahn-Banach, if Y is a closed subspace of X and $x_0 \notin Y$, then:
(A) $x_0 \in Y^\perp$
(B) $\exists f \in X^*$ with $f|_Y = 0$ and $f(x_0) \neq 0$
(C) Y is not closed
(D) $x_0 = 0$

Answer: (B)

Explanation: *HB separation from a subspace: if $x_0 \notin Y$ (closed), then $\operatorname{dist}(x_0, Y) > 0$; define f_0 on $Y \oplus \operatorname{span}\{x_0\}$ by $f_0(y + \alpha x_0) = \alpha \operatorname{dist}(x_0, Y)$; extend by HB. This $F \in X^*$ vanishes on Y and $F(x_0) = \operatorname{dist}(x_0, Y) \neq 0$.*

Q81

A linear functional on a finite-dimensional space is:
(A) Bounded only if the space is $\mathbb{R}$
(B) Always bounded (continuous)
(C) Bounded only if it is injective
(D) Never bounded

Answer: (B)

Explanation: *On a finite-dimensional normed space, every linear map (including linear functionals) is bounded (continuous). This is because all norms are equivalent in finite dimensions, and linear maps are bounded in any finite-dimensional space.*

Q82

The **evaluation functional** $\delta_t : C[a, b] \to \mathbb{R}$ defined by $\delta_t(f) = f(t)$ is:
(A) Unbounded
(B) Bounded, with $\|\delta_t\| = 1$
(C) Not linear
(D) Bounded, with $\|\delta_t\| = \infty$

Answer: (B)

Explanation: *δ_t is bounded: $|\delta_t(f)| = |f(t)| \le \|f\|_\infty$, so $\|\delta_t\| \le 1$. For $f \equiv 1$: $|\delta_t(f)| = 1 = \|f\|_\infty$, so $\|\delta_t\| = 1$. This is the Dirac delta at t (continuous version).*

Q83

The HB theorem is used to show that for any normed X:
(A) $X^* = \{0\}$
(B) X^* has enough functionals to separate points of X
(C) X^* is finite-dimensional

(D) X is reflexive

Answer: (B)

Explanation: *HB ensures X^* "separates points": for $x \neq y$ in X, $\exists f \in X^*$ with $f(x) \neq f(y)$. Proof: $x - y \neq 0$; by HB, $\exists f$ with $f(x-y) \neq 0$, i.e., $f(x) \neq f(y)$.*

Q84

A closed hyperplane in X is a set of the form:

(A) A closed subspace of codimension 1
(B) $\{x : f(x) = \gamma\}$ for $f \in X^*$, $\gamma \in \mathbb{K}$
(C) The kernel of a bounded functional
(D) All of the above

Answer: (D)

Explanation: *All three descriptions are equivalent. A hyperplane is $f^{-1}(\gamma) = \{x : f(x) = \gamma\}$ for some non-zero $f \in X^*$. For $\gamma = 0$: it equals $\ker f$ (a closed codimension-1 subspace). For $\gamma \neq 0$: an affine subspace. The separation theorem uses such hyperplanes.*

Q85

HB theorem requires the space to be over:

(A) $\mathbb{R}$ only
(B) $\mathbb{C}$ only
(C) $\mathbb{R}$ or $\mathbb{C}$ (any field of scalars)
(D) A field of characteristic 0 other than $\mathbb{R}$ or $\mathbb{C}$

Answer: (C)

Explanation: *HB holds over $\mathbb{R}$ (real version, proved using Zorn's lemma) and over $\mathbb{C}$ (complex version, derived from the real version using Bohnenblust-Sobczyk). For the complex case, one uses $\Re f$ (real part) and extends, then sets $F(x) = \overline{f_{\mathbb{R}}}(ix) - i f_{\mathbb{R}}(x)$.*

Q86

The dual of $C[a,b]$ (with sup norm) is:

(A) $C[a,b]$ itself
(B) $L^1[a,b]$
(C) The space of regular Borel measures on $[a,b]$ (Riesz-Markov)
(D) $L^\infty[a,b]$

Answer: (C)

Explanation: *By the Riesz-Markov representation theorem: $(C[a,b])^*$ is isometrically isomorphic to the space of finite regular Borel signed measures on $[a,b]$. Every bounded linear functional on $C[a,b]$ has the form $f(g) = \int g\,d\mu$ for a unique such measure μ.*

Q87

The canonical map $J : X \to X^{**}$ defined by $Jx(f) = f(x)$ is:

(A) Always surjective
(B) An isometric linear injection
(C) Surjective only when X is Hilbert
(D) Not always linear

Answer: (B)

Explanation: *J is always an isometric linear injection. $\|Jx\| = \|x\|$ (by HB: $\|Jx\| = \sup_{\|f\|\le 1} |f(x)| = \|x\|$). J is surjective $\iff$ X is reflexive (which includes all Hilbert spaces and ℓ^p, $1 < p < \infty$, but not all Banach spaces).*

Q88

If X is a normed space and $A \subseteq X$ is a convex set, its **weak closure** equals:
(A) Its norm closure
(B) X itself
(C) Its interior
(D) A if it is already weakly closed

Answer: (A)

Explanation: *For convex sets in a normed space: norm closure = weak closure. This is a deep consequence of HB: a convex set is closed iff it is weakly closed. So for convex sets, two apparently different notions of closure coincide.*

Q89

The HB theorem is NOT needed to show:
(A) The dual of $\mathbb{R}^n$ is $\mathbb{R}^n$
(B) Nontrivial functionals on infinite-dimensional spaces exist
(C) The bidual embedding is isometric
(D) The dual space is complete

Answer: (A)

Explanation: *On $\mathbb{R}^n$ (finite-dimensional), linear functionals always exist and are always bounded (continuous) — no HB needed. The dual $(\mathbb{R}^n)^* \cong \mathbb{R}^n$ follows from basic linear algebra. HB is essential for infinite-dimensional spaces.*

Q90

$(\ell^1)^* \cong$:
(A) ℓ^1
(B) ℓ^2
(C) ℓ^∞
(D) c_0

Answer: (C)

Explanation: *$(\ell^1)^* \cong \ell^\infty$. Every bounded linear functional on ℓ^1 is of the form $f(x) = \sum x_n y_n$ for $(y_n) \in \ell^\infty$, and $\|f\| = \|y\|_\infty$. This is a classical duality: the conjugate exponent of $p = 1$ is $q = \infty$.*

[DAY] DAY 8 — The Big Three Theorems ★★ (Most Asked)

Topics: Uniform Boundedness Principle (Banach-Steinhaus), Open Mapping Theorem, Closed Graph Theorem, consequences and applications.

4. The Uniform Boundedness Principle (Banach-Steinhaus)

[STAR] Uniform Boundedness Principle (UBP)

Let X be a **Banach space**, Y a normed space, and $\{T_\alpha\}_{\alpha\in\Lambda} \subseteq \mathcal{B}(X, Y)$ a family of bounded linear operators. If the family is **pointwise bounded**:

$$\forall\, x \in X, \quad \sup_{\alpha} \|T_\alpha x\| < \infty,$$

then the family is **uniformly bounded**:

$$\sup_{\alpha} \|T_\alpha\| < \infty.$$

Consequence: if $\sup_\alpha \|T_\alpha\| = \infty$*, there exists* $x \in X$ *with* $\sup_\alpha \|T_\alpha x\| = \infty$*.*

[KEY] UBP: Intuition and Application

Intuition: Pointwise bounded (bounded at each point) $\Rightarrow$ uniformly bounded (bounded as operators). "You can't be small everywhere but big collectively."

Application: Divergence of Fourier series. For each $f \in C[0, 1]$, define $S_n f = n$-th partial Fourier sum. Then $\|S_n\| \to \infty$ (proved by finding specific f). UBP then gives: $\exists\, f \in C[0, 1]$ with $S_n f(x_0) \not\to f(x_0)$ for some x_0. So NOT every continuous function has a pointwise-convergent Fourier series.

5. Open Mapping Theorem

[STAR] Open Mapping Theorem (OMT)

Let X and Y be **Banach spaces** and $T : X \to Y$ a **surjective** bounded linear operator. Then T is an **open mapping**: T maps open sets to open sets.

$$T \text{ surjective, bounded, linear, } X, Y \text{ Banach} \implies T \text{ open.}$$

[COR] Corollaries of OMT

(i) **Bounded Inverse Theorem:** If $T : X \to Y$ is a bijective bounded linear operator between Banach spaces, then $T^{-1} : Y \to X$ is also bounded (continuous).

(ii) If $T : X \to Y$ is bounded, linear, bijective between Banach spaces, then T is a **homeomorphism** (and a topological isomorphism).

(iii) Two Banach space norms on X with one finer than the other must be equivalent.

6. Closed Graph Theorem

[STAR] Closed Graph Theorem (CGT)

Let X and Y be **Banach spaces** and $T : X \to Y$ a linear operator. T is **bounded** (continuous) $\iff$ its **graph** is closed:

$$G(T) = \{(x, Tx) : x \in X\} \subseteq X \times Y \text{ is closed.}$$

Equivalently: T is bounded $\iff$ ($x_n \to x$ and $Tx_n \to y \Rightarrow y = Tx$).

[KEY] Three Big Theorems: Quick Summary

Theorem	Statement	Key Requirement
UBP (Banach-Steinhaus)	Pointwise bounded $\Rightarrow$ Uniformly bounded	X Banach
Open Mapping	Surjective bounded $\Rightarrow$ Open map	X, Y both Banach
Closed Graph	Graph closed $\iff$ Operator bounded	X, Y both Banach

All three follow from BCT (Baire Category Theorem)!

Day 8 MCQs — The Big Three Theorems

Q91

The **Uniform Boundedness Principle** requires X to be:

(A) Finite-dimensional
(B) A Banach space (complete normed space)
(C) A Hilbert space
(D) Separable

Answer: (B)

Explanation: *UBP requires X to be a Banach space. Completeness is essential (Baire Category Theorem is used in the proof). Without completeness, the conclusion fails: there exist pointwise bounded families on incomplete spaces that are not uniformly bounded.*

Q92

If $\{T_n\}$ is a sequence of bounded operators on Banach X and $\sup_n \|T_n x\| < \infty$ for each x, then:

(A) $T_n \to 0$ uniformly
(B) $\sup_n \|T_n\| < \infty$
(C) Each $T_n = 0$
(D) $\sup_n \|T_n\| = \infty$

Answer: (B)

Explanation: *By UBP: pointwise boundedness ($\sup_n \|T_n x\| < \infty$ for each x) on a Banach space X implies uniform boundedness ($\sup_n \|T_n\| < \infty$). This is the key conclusion of Banach-Steinhaus.*

Q93

UBP is proved using:

(A) Hahn-Banach theorem
(B) Open Mapping Theorem
(C) Baire Category Theorem
(D) Cauchy-Schwarz inequality

Answer: (C)

Explanation: *All three Big Theorems are proved using BCT. For UBP: define $F_n = \{x : \sup_k \|T_k x\| \leq n\}$ (closed sets by continuity). If $\sup_k \|T_k\| = \infty$, then $X = \bigcup F_n$ implies by BCT that some F_n has non-empty interior — leading to uniform boundedness.*

Q94

The **Open Mapping Theorem** states: if $T : X \to Y$ is a surjective bounded linear map between Banach spaces, then:

(A) T is an isometry
(B) T maps open sets to open sets
(C) T has a bounded right inverse
(D) T is compact

Answer: (B)

Explanation: *OMT: surjective bounded linear T between Banach spaces $\Rightarrow$ T is open (open sets map to open sets). Consequence: if T is also injective (bijective), then T^{-1} is bounded (Bounded Inverse Theorem).*

Q95

The **Bounded Inverse Theorem** states: if $T : X \to Y$ is bijective bounded linear between Banach spaces, then:

(A) T^{-1} is linear but not necessarily bounded
(B) T^{-1} is bounded (continuous)
(C) T must be an isometry
(D) T^{-1} does not exist

Answer: (B)

Explanation: *Bounded Inverse Theorem (corollary of OMT): bijective bounded linear T between Banach spaces has bounded inverse T^{-1}. Key: both spaces must be Banach and T must be surjective. Without surjectivity or completeness, the inverse may be unbounded.*

Q96

The **Closed Graph Theorem** states: $T : X \to Y$ (Banach spaces) is bounded $\iff$:

(A) T has a bounded right inverse
(B) The graph $G(T) = \{(x, Tx)\}$ is closed in $X \times Y$
(C) $\ker T$ is closed
(D) T is surjective

Answer: (B)

Explanation: *CGT: for linear T between Banach spaces, bounded $\iff$ graph $G(T) = \{(x, Tx) : x \in X\}$ is a closed subset of $X \times Y$. This provides an easier way to prove boundedness: just show that if $x_n \to x$ and $Tx_n \to y$ then $y = Tx$.*

Q97

The CGT is equivalent to: $T : X \to Y$ is bounded iff ($x_n \to x$ and $Tx_n \to y$) $\Rightarrow$:

(A) $y = 0$
(B) $y = Tx$
(C) $x = 0$
(D) T is injective

Answer: (B)

Explanation: *Closed graph condition: $G(T)$ closed $\iff$ the limit y of Tx_n equals Tx whenever $x_n \to x$. This is precisely the "sequential" characterization of $G(T)$ being closed in $X \times Y$.*

Q98

All three Big Theorems (UBP, OMT, CGT) are consequences of:

(A) Hahn-Banach Theorem
(B) Baire Category Theorem
(C) Cauchy-Schwarz Inequality
(D) Riesz Representation Theorem

Answer: (B)

Explanation: *All three follow from BCT. BCT says a complete metric space is of second category. Applied to Banach spaces (which are complete metric spaces), it gives: X cannot be written as a countable union of nowhere dense sets. This is the key ingredient in proofs of UBP, OMT, and CGT.*

Q99

Which of the following is an application of UBP?

(A) Existence of a Hilbert basis
(B) Existence of a continuous function with divergent Fourier series
(C) Hahn-Banach extension
(D) Riesz decomposition

Answer: (B)

Explanation: *Classic UBP application: the partial Fourier sum operators $S_n : C[0, 1] \to \mathbb{R}$ have $\|S_n\| \to \infty$. By UBP (contrapositive): $\exists\, f \in C[0, 1]$ with $\sup_n |S_n f(0)| = \infty$, i.e., the Fourier series of f diverges at 0.*

Q100

The OMT requires T to be:

(A) Injective only
(B) Surjective (onto)
(C) An isometry
(D) Self-adjoint

Answer: (B)

Explanation: *OMT requires T to be surjective (onto). Without surjectivity, T need not be open (e.g., $T : \mathbb{R} \to \mathbb{R}^2$, $Tx = (x, 0)$ is injective but not open). Both domain and codomain must be Banach spaces.*

Q101

If X is a Banach space with two norms $\|\cdot\|_1$ and $\|\cdot\|_2$, with $\|\cdot\|_2$ stronger (i.e., $\|x\|_1 \le C\|x\|_2$), and both make X Banach, then by OMT:

(A) The norms are equal

(B) The norms are equivalent
(C) $\|\cdot\|_2$ must be the sup norm
(D) The space must be finite-dimensional

Answer: (B)

Explanation: *If both norms make X Banach and one is stronger, by OMT (applied to the identity map $I : (X, \|\cdot\|_2) \to (X, \|\cdot\|_1)$, which is bounded surjective bijective between Banach spaces) the inverse is bounded, giving $\|x\|_2 \le C'\|x\|_1$. So the norms are equivalent.*

Q102

A linear map $T : X \to Y$ between normed spaces is **bounded** if:
(A) T is bounded on bounded sets
(B) $\|T\| = \sup_{\|x\|\le 1} \|Tx\| < \infty$
(C) T maps the unit ball to a compact set
(D) T has a bounded right inverse

Answer: (B)

Explanation: *A linear map is bounded if its operator norm $\|T\| = \sup_{\|x\|\le 1} \|Tx\| < \infty$. Equivalently: $\exists C > 0$ s.t. $\|Tx\| \le C\|x\|$ for all x. For linear maps: bounded $\iff$ continuous (at any point $\iff$ everywhere).*

Q103

UBP gives: if $T_n : X \to Y$ (Banach) and $T_n x \to Tx$ for all x, then:
(A) T is compact
(B) $\sup_n \|T_n\| < \infty$ and T is bounded
(C) $\|T_n - T\| \to 0$ (uniform convergence)
(D) T is surjective

Answer: (B)

Explanation: *If $T_n x \to Tx$ for all x (pointwise/strong convergence), then by UBP: $\sup_n \|T_n\| < \infty$ (since pointwise bounded). The limit T is bounded with $\|T\| \le \liminf \|T_n\|$. Note: strong convergence does NOT imply uniform convergence in general.*

Q104

The CGT is used to show that a symmetric operator on a Hilbert space is bounded if:
(A) It is compact
(B) It has a closed graph (which symmetric operators automatically have)
(C) It is self-adjoint
(D) It is positive

Answer: (B)

Explanation: *The Hellinger-Toeplitz theorem: a symmetric operator $T : H \to H$ defined everywhere on a Hilbert space must be bounded. Proof via CGT: T is symmetric so $\langle Tx_n, y\rangle \to \langle x, Ty\rangle$; if $x_n \to x$ and $Tx_n \to z$, then for all y: $\langle z, y\rangle = \langle Tx, y\rangle$, so $z = Tx$. Graph is closed, hence T is bounded by CGT.*

Q105

An operator $T : X \to Y$ is **open** if:
(A) T is continuous
(B) T maps every open set in X to an open set in Y

(C) T has open kernel
(D) T maps closed sets to closed sets

Answer: (B)

Explanation: *An open map sends open sets to open sets. Note: continuous maps send open sets to preimages that are open (the other direction). Open and continuous are independent in general; OMT says surjective bounded linear between Banach $\Rightarrow$ open.*

Q106

Which is the correct implication for bounded linear $T : X \to Y$ between Banach spaces?
(A) Bijective $\Rightarrow$ isometric
(B) Bijective $\Rightarrow$ T^{-1} is bounded
(C) Surjective $\Rightarrow$ injective
(D) Injective $\Rightarrow$ surjective

Answer: (B)

Explanation: *By the Bounded Inverse Theorem (corollary of OMT): if T is bijective bounded linear between Banach spaces, then T^{-1} is also bounded. This is remarkable: algebraic bijectivity + continuity of T forces continuity of T^{-1}.*

Q107

The conclusion of UBP can fail if X is:
(A) A Hilbert space
(B) An incomplete normed space
(C) Infinite-dimensional
(D) Separable

Answer: (B)

Explanation: *UBP can fail for incomplete spaces. Counterexample: on the space c_{00} (finitely supported sequences) with ℓ^2 norm (not complete), define $T_n(e_k) = ke_k$ for $k \le n$, 0 otherwise. Then $\sup_n \|T_n x\| < \infty$ for each $x \in c_{00}$ but $\sup_n \|T_n\| = \infty$.*

Q108

The OMT implies that a surjective bounded linear operator $T : X \to Y$ between Banach spaces satisfies:
(A) $T(B(0,1)) \supseteq B(0,\delta)$ for some $\delta > 0$
(B) $T(B(0,1)) = B(0,1)$
(C) T is an isometry
(D) $T(B(0,1))$ is compact

Answer: (A)

Explanation: *OMT precisely means: T maps open sets to open sets. So $T(B(0,1))$ is open (since $B(0,1)$ is open) and contains 0 (since $T(0) = 0$). Hence $T(B(0,1)) \supseteq B(0,\delta)$ for some $\delta > 0$. This δ relates to the boundedness of T^{-1}.*

Q109

Which theorem is most directly used to prove: "If $T_n \in \mathcal{B}(X,Y)$ and $T_n x \to Tx$ for all x, then T is bounded"?
(A) Hahn-Banach

(B) Closed Graph Theorem
(C) Uniform Boundedness Principle
(D) Open Mapping Theorem

Answer: (C)

Explanation: *UBP is used: strong convergence $T_n x \to Tx$ means $\sup_n \|T_n x\| < \infty$ for each fixed x (since convergent sequences are bounded). By UBP: $\sup_n \|T_n\| = C < \infty$. Then $\|Tx\| = \lim \|T_n x\| \leq C\|x\|$, so T is bounded.*

Q110

The CGT requires BOTH X and Y to be:
(A) Finite-dimensional
(B) Hilbert spaces
(C) Banach spaces
(D) Separable

Answer: (C)

Explanation: *CGT requires both X and Y to be Banach spaces. Without completeness, the theorem fails. Example of failure: $T = \frac{d}{dt} : C^1[0,1] \to C[0,1]$ (with sup norm) has closed graph but is not bounded when the domain is not taken to be complete.*

[DAY] DAY 9 — Operators & Spectral Theory

Topics: Bounded linear operators, compact operators, adjoint operators, self-adjoint/unitary/normal operators, spectrum (point/continuous/residual), spectral theorem.

7. Bounded Linear Operators

[DEF] Bounded Linear Operators

$\mathcal{B}(X,Y) = \{T : X \to Y : T \text{ linear, bounded}\}$ with operator norm $\|T\| = \sup_{\|x\|\leq 1} \|Tx\|$. If $X = Y$: write $\mathcal{B}(X) = \mathcal{B}(X,X)$. $(\mathcal{B}(X,Y), \|\cdot\|)$ is a normed space; Banach if Y is Banach. $\mathcal{B}(X)$ is also an algebra under composition: $\|ST\| \leq \|S\|\|T\|$.

[DEF] Compact Operator

$T : X \to Y$ is **compact** if for every bounded set $A \subseteq X$, the image $T(A)$ is **precompact** (its closure $\overline{T(A)}$ is compact). Equivalently: T maps bounded sequences to sequences with convergent subsequences.

[THM] Properties of Compact Operators

(i) Every compact operator is bounded.

(ii) Compact operators form a closed subspace of $\mathcal{B}(X,Y)$.

(iii) Composition of bounded and compact is compact: ST compact if S or T is compact.

(iv) Every finite-rank operator (finite-dimensional range) is compact.

(v) In Hilbert spaces: T compact $\iff$ $T = \lim \|T_n - T\| = 0$ where each T_n is finite-rank.

(vi) Identity $I : X \to X$ is compact $\iff$ X is finite-dimensional.

[DEF] Adjoint Operator

For $T \in \mathcal{B}(H,K)$ (Hilbert spaces), the **adjoint** $T^* \in \mathcal{B}(K,H)$ is defined by:

$$\langle Tx,\, y\rangle = \langle x,\, T^*y\rangle \quad \forall\, x \in H,\, y \in K.$$

Properties: $\|T^*\| = \|T\|$, $\|T^*T\| = \|T\|^2$, $(T^*)^* = T$, $(ST)^* = T^*S^*$.

[DEF] Special Operators

- **Self-adjoint** (Hermitian): $T^* = T$ (i.e., $\langle Tx,\, y\rangle = \langle x,\, Ty\rangle$ for all x, y).
- **Unitary** (on Hilbert): $TT^* = T^*T = I$ (equivalently: T is an isometric isomorphism).
- **Normal**: $TT^* = T^*T$ (commutes with its adjoint).
- **Positive**: $\langle Tx,\, x\rangle \geq 0$ for all x.
- **Isometry**: $\|Tx\| = \|x\|$ for all x (equivalently $T^*T = I$).

8. Spectral Theory

[DEF] Spectrum and Resolvent

For $T \in \mathcal{B}(X)$ (Banach), the **resolvent set** and **spectrum** are:

$$\rho(T) = \{\lambda \in \mathbb{C} : (\lambda I - T)^{-1} \in \mathcal{B}(X)\} \quad \text{(resolvent set)}$$
$$\sigma(T) = \mathbb{C} \setminus \rho(T) = \{\lambda : (\lambda I - T) \text{ not invertible in } \mathcal{B}(X)\}$$

The spectrum decomposes as:

$$\sigma(T) = \sigma_p(T) \cup \sigma_c(T) \cup \sigma_r(T).$$

[DEF] Point, Continuous, and Residual Spectrum

- **Point spectrum** $\sigma_p(T)$: λ s.t. $\ker(\lambda I - T) \neq \{0\}$ (i.e., λ is an **eigenvalue**: $\exists x \neq 0$ with $Tx = \lambda x$).
- **Continuous spectrum** $\sigma_c(T)$: λ s.t. $(\lambda I - T)$ is injective, $\operatorname{ran}(\lambda I - T)$ is dense in X, but $(\lambda I - T)^{-1}$ is unbounded.
- **Residual spectrum** $\sigma_r(T)$: λ s.t. $(\lambda I - T)$ is injective but $\operatorname{ran}(\lambda I - T)$ is NOT dense in X.

[THM] Key Facts About the Spectrum

(i) $\sigma(T)$ is a non-empty compact subset of $\mathbb{C}$ with $\sigma(T) \subseteq \bar{B}(0, \|T\|)$.

(ii) The **spectral radius**: $r(T) = \sup\{|\lambda| : \lambda \in \sigma(T)\} = \lim \|T^n\|^{1/n}$.

(iii) For self-adjoint T on Hilbert H: $\sigma(T) \subseteq \mathbb{R}$ and $\sigma_r(T) = \emptyset$.

(iv) For unitary T: $\sigma(T) \subseteq \{\lambda : |\lambda| = 1\}$ (on the unit circle).

(v) For compact T on infinite-dim. Banach: $0 \in \sigma(T)$; nonzero spectrum = eigenvalues with finite multiplicity, accumulating only at 0.

[THM] Spectral Theorem (Self-Adjoint Compact Operators)

Let $T : H \to H$ be a **compact self-adjoint** operator on a Hilbert space $H \neq \{0\}$. Then:

(i) T has a **real** eigenvalue λ_1 with $|\lambda_1| = \|T\|$.

(ii) There exists a complete orthonormal system $\{e_n\}$ of eigenvectors of T.

(iii) $T = \sum_{n=1}^{\infty} \lambda_n \langle \cdot, e_n \rangle e_n$ (spectral decomposition).

(iv) Eigenvalues $\lambda_n \to 0$ (if H is infinite-dimensional).

Day 9 MCQs — Operators & Spectral Theory

Q111

A bounded linear operator $T : X \to Y$ is **compact** if:

(A) T maps bounded sets to bounded sets
(B) T maps bounded sets to precompact (relatively compact) sets
(C) T is a finite-rank operator
(D) $\ker T = \{0\}$

Answer: (B)

Explanation: *Compact operator: every bounded set maps to a precompact set (closure is compact). Equivalently: bounded sequences map to sequences with convergent subsequences. (C) is sufficient but not necessary: compact operators may have infinite rank (e.g., diagonal operators with $\lambda_n \to 0$ on ℓ^2).*

Q112

The **adjoint** T^* of $T \in \mathcal{B}(H)$ is defined by:
(A) $T^* = T^{-1}$
(B) $\langle Tx, y\rangle = \langle x, T^*y\rangle$ for all $x, y \in H$
(C) $T^*x = \overline{Tx}$
(D) T^* is the transpose of the matrix of T

Answer: (B)

Explanation: *The adjoint T^* is defined by $\langle Tx, y\rangle = \langle x, T^*y\rangle$. By Riesz representation theorem, T^* exists uniquely. Properties: $\|T^*\| = \|T\|$, $(T^*)^* = T$, $(ST)^* = T^*S^*$.*

Q113

T is **self-adjoint** if:
(A) $T = T^{-1}$
(B) $T^* = T$
(C) $TT^* = I$
(D) $\|Tx\| = \|x\|$ for all x

Answer: (B)

Explanation: *Self-adjoint (Hermitian): $T^* = T$, i.e., $\langle Tx, y\rangle = \langle x, Ty\rangle$ for all x, y. Key properties: all eigenvalues are real; eigenvectors for different eigenvalues are orthogonal; $\sigma(T) \subseteq \mathbb{R}$; $\sigma_r(T) = \emptyset$.*

Q114

A **unitary** operator $U : H \to H$ satisfies:
(A) $U^* = U$
(B) $UU^* = U^*U = I$
(C) $\|Ux\| = \|x\|$ and U is surjective
(D) Both (B) and (C)

Answer: (D)

Explanation: *Unitary: $UU^* = U^*U = I$, which is equivalent to: U is an isometric isomorphism (surjective isometry). The spectrum of a unitary lies on the unit circle $\{|\lambda| = 1\}$.*

Q115

The **spectrum** $\sigma(T)$ of $T \in \mathcal{B}(X)$ is:
(A) The set of eigenvalues of T
(B) The set of $\lambda \in \mathbb{C}$ for which $(\lambda I - T)$ is not invertible in $\mathcal{B}(X)$
(C) The range of T
(D) The set of λ for which $(\lambda I - T)$ is compact

Answer: (B)

Explanation: *Spectrum: $\sigma(T) = \{\lambda \in \mathbb{C} : (\lambda I - T) \notin GL(\mathcal{B}(X))\}$. This includes eigenvalues (point spectrum) but also λ's where $\lambda I - T$ is injective but not surjective or the inverse is unbounded. $\sigma(T) \neq$ eigenvalues in infinite dimensions!*

Q116

λ is an **eigenvalue** of T if:
(A) $Tx = \lambda x$ for all x
(B) $\exists\, x \neq 0$ with $Tx = \lambda x$
(C) $(\lambda I - T)$ is surjective
(D) $\lambda \in \rho(T)$

Answer: (B)

Explanation: *Eigenvalue: $\exists x \neq 0$ with $Tx = \lambda x$ (i.e., $\ker(\lambda I - T) \neq \{0\}$). Such x is an eigenvector. Eigenvalues are in the point spectrum $\sigma_p(T) \subseteq \sigma(T)$. In finite dimensions, $\sigma(T) = \sigma_p(T)$ (only eigenvalues).*

Q117

For a self-adjoint operator T on Hilbert H, the spectrum satisfies:
(A) $\sigma(T) \subseteq i\mathbb{R}$ (purely imaginary)
(B) $\sigma(T) \subseteq \mathbb{R}$ (real)
(C) $\sigma(T)$ is finite
(D) $\sigma(T) \subseteq \{|\lambda| = 1\}$

Answer: (B)

Explanation: *For self-adjoint T: $\sigma(T) \subseteq \mathbb{R}$. Proof: if $\lambda = \alpha + i\beta$ with $\beta \neq 0$ and $Tx = \lambda x$, then $\langle Tx, x\rangle = \lambda\|x\|^2$ but also $\langle Tx, x\rangle = \langle x, Tx\rangle = \bar{\lambda}\|x\|^2$, giving $\lambda = \bar{\lambda}$, so $\beta = 0$.*

Q118

The spectral radius formula is:
(A) $r(T) = \|T\|$
(B) $r(T) = \lim_{n\to\infty} \|T^n\|^{1/n}$
(C) $r(T) = \|\sigma(T)\|$
(D) $r(T) = \|T^2\|^{1/2}$

Answer: (B)

Explanation: *Spectral radius formula (Gelfand): $r(T) = \sup\{|\lambda| : \lambda \in \sigma(T)\} = \lim_{n\to\infty} \|T^n\|^{1/n}$. In general $r(T) \leq \|T\|$. For normal operators on Hilbert: $r(T) = \|T\|$.*

Q119

The spectrum of a compact operator on an infinite-dimensional Banach space:
(A) Is always empty
(B) Consists of finitely many nonzero eigenvalues and 0
(C) Contains 0; nonzero elements are eigenvalues with finite multiplicity, accumulating only at 0
(D) Is uncountable

Answer: (C)

Explanation: *For compact T on infinite-dim. X: $0 \in \sigma(T)$ always; every $\lambda \neq 0$ in $\sigma(T)$ is an eigenvalue (point spectrum) with finite-dimensional eigenspace; and the nonzero eigenvalues can only accumulate at 0 (Riesz-Schauder theory).*

Q120

The identity operator $I : X \to X$ is compact iff:

(A) X is a Hilbert space
(B) X is finite-dimensional
(C) X is separable
(D) X is reflexive

Answer: (B)

Explanation: *I compact $\iff$ the unit ball $B(0,1) = I(B(0,1))$ is precompact $\iff$ the unit ball is compact $\iff$ X finite-dimensional (Riesz). In infinite dimensions, the unit ball is never compact, so I is never compact.*

Q121

A normal operator satisfies:

(A) $T^* = T$
(B) $TT^* = T^*T$
(C) $TT^* = I$
(D) $T^2 = T$

Answer: (B)

Explanation: *Normal: $TT^* = T^*T$. Normal operators include: self-adjoint ($T^* = T$), unitary ($TT^* = T^*T = I$), skew-adjoint ($T^* = -T$). The spectral theorem holds in its most general form for normal operators on Hilbert spaces.*

Q122

The **residual spectrum** $\sigma_r(T)$ consists of λ where:

(A) $(\lambda I - T)$ is not injective
(B) $(\lambda I - T)$ is injective but $\operatorname{ran}(\lambda I - T)$ is not dense
(C) $(\lambda I - T)$ is surjective but not injective
(D) $(\lambda I - T)^{-1}$ is bounded

Answer: (B)

Explanation: *Residual spectrum: λ where $(\lambda I - T)$ is injective (no eigenvalue) but the range is not dense. For self-adjoint operators on Hilbert: $\sigma_r(T) = \emptyset$ (eigenspaces of distinct eigenvalues are orthogonal and span H). Residual spectrum is harder to "see."*

Q123

Every finite-rank operator (finite-dimensional range) is:

(A) Unitary
(B) Self-adjoint
(C) Compact
(D) Invertible

Answer: (C)

Explanation: *Finite-rank $\Rightarrow$ compact: if $\operatorname{ran} T$ is finite-dimensional, then $T(B(0,1)) \subseteq \operatorname{ran} T$ is bounded in a finite-dimensional space, hence precompact. Compact operators can be approximated in norm by finite-rank operators (in Hilbert spaces).*

Q124

For a compact self-adjoint operator T on Hilbert H, the spectral theorem says:

(A) T has exactly one eigenvalue
(B) T has a complete ONS of eigenvectors with real eigenvalues $\lambda_n \to 0$
(C) $\sigma(T) = \mathbb{R}$
(D) T is unitary

Answer: (B)

Explanation: *Spectral theorem for compact self-adjoint T: $\exists$ CONS $\{e_n\}$ of eigenvectors with $Te_n = \lambda_n e_n$, $\lambda_n \in \mathbb{R}$, $\lambda_n \to 0$. The expansion $T = \sum \lambda_n \langle \cdot, e_n \rangle e_n$ holds. This is the infinite-dimensional analogue of matrix diagonalization.*

Q125

For isometry T (i.e., $\|Tx\| = \|x\|$ for all x), we have:

(A) $T^*T = I$
(B) $TT^* = I$
(C) $T = T^*$
(D) $\sigma(T) = \{1\}$

Answer: (A)

Explanation: *Isometry: $\|Tx\|^2 = \|x\|^2$ for all x, i.e., $\langle Tx, Tx \rangle = \langle x, x \rangle$, i.e., $\langle T^*Tx, x \rangle = \langle x, x \rangle$. So $T^*T = I$. Note: $TT^* = I$ requires additionally that T is surjective (making it unitary). An isometry is always injective; if also surjective, it is unitary.*

Q126

The **resolvent** $R(\lambda, T) = (\lambda I - T)^{-1}$ exists and is bounded when $\lambda \in$:

(A) $\sigma(T)$
(B) $\rho(T)$ (the resolvent set)
(C) $\sigma_p(T)$
(D) The unit circle

Answer: (B)

Explanation: *The resolvent $R(\lambda, T)$ exists and is bounded ($\in \mathcal{B}(X)$) iff $\lambda \in \rho(T)$ (the resolvent set = complement of spectrum). The spectrum is where the resolvent fails to exist or is unbounded.*

Q127

The spectrum $\sigma(T)$ is always:

(A) Open
(B) Unbounded
(C) A compact subset of $\mathbb{C}$ with $\sigma(T) \subseteq \bar{B}(0, \|T\|)$
(D) Contained in the real line

Answer: (C)

Explanation: *$\sigma(T)$ is always compact (closed and bounded). Bounded: $\|T\| < |\lambda|$ implies $(\lambda I - T)^{-1}$ exists (Neumann series), so $|\lambda| > \|T\| \Rightarrow \lambda \in \rho(T)$. Closed: resolvent set is open. Non-empty: by Liouville's theorem. (D) holds only for self-adjoint operators.*

Q128

For $T \in \mathcal{B}(X)$, if $|\lambda| > \|T\|$, then $\lambda \in \rho(T)$ and:

(A) $R(\lambda, T) = (\lambda I - T)^{-1} = \sum_{n=0}^{\infty} \lambda^{-n-1} T^n$ (Neumann series)
(B) $R(\lambda, T) = T^{-1}$
(C) $R(\lambda, T)$ does not exist
(D) λ is an eigenvalue

Answer: (A)

Explanation: *Neumann series: for $|\lambda| > \|T\|$, $(\lambda I - T)^{-1} = \frac{1}{\lambda} \sum_{n=0}^{\infty} (\frac{T}{\lambda})^n = \sum_{n=0}^{\infty} \frac{T^n}{\lambda^{n+1}}$, convergent since $\|T/\lambda\| < 1$. This is the resolvent formula outside the disk $\|T\|$.*

Q129

Eigenvectors of a self-adjoint operator corresponding to **distinct** eigenvalues are:

(A) Parallel (proportional)
(B) Orthogonal
(C) Linearly dependent
(D) Identical

Answer: (B)

Explanation: *If $Tu = \lambda u$ and $Tv = \mu v$ with $\lambda \neq \mu$ (both real for self-adjoint), then $\lambda\langle u, v\rangle = \langle Tu, v\rangle = \langle u, Tv\rangle = \bar{\mu}\langle u, v\rangle = \mu\langle u, v\rangle$ (since μ real). So $(\lambda - \mu)\langle u, v\rangle = 0$. Since $\lambda \neq \mu$: $\langle u, v\rangle = 0$, i.e., $u \perp v$.*

Q130

The **point spectrum** of the left-shift operator $L : \ell^2 \to \ell^2$ ($(Lx)_n = x_{n+1}$) is:

(A) Empty
(B) $\{\lambda : |\lambda| < 1\}$
(C) $\{\lambda : |\lambda| = 1\}$
(D) $\{\lambda : |\lambda| > 1\}$

Answer: (B)

Explanation: *For left-shift L: $Lx = \lambda x \Rightarrow x_{n+1} = \lambda x_n \Rightarrow x_n = x_1 \lambda^{n-1}$. For $(x_n) \in \ell^2$: need $\sum |\lambda|^{2(n-1)} < \infty$, i.e., $|\lambda| < 1$. So $\sigma_p(L) = \{\lambda : |\lambda| < 1\}$. The full spectrum is $\{|\lambda| \leq 1\}$.*

Q131

The space $\mathcal{B}(X, Y)$ of bounded linear operators, with operator norm, is:

(A) A Banach space only if X is Banach
(B) A Banach space if and only if Y is Banach
(C) Always a Banach space
(D) A Hilbert space if $X = Y$

Answer: (B)

Explanation: *$\mathcal{B}(X, Y)$ is a Banach space iff Y is Banach. The completeness of the codomain ensures Cauchy sequences of operators converge to bounded operators. The completeness of X is NOT needed (only Y matters).*

Q132

A projection $P : H \to H$ is **orthogonal** iff:

(A) $P = P^{-1}$

(B) $P = P^2$ and $P = P^*$
(C) $P = 0$ or $P = I$
(D) $\|P\| > 1$

Answer: (B)

Explanation: *Orthogonal projection: $P^2 = P$ (idempotent) AND $P^* = P$ (self-adjoint). A general projection only requires $P^2 = P$. An orthogonal projection is the unique projection onto a closed subspace with minimum norm ($\|P\| \leq 1$, and $\|P\| = 1$ if $P \neq 0$).*

Q133

Eigenvalues of a **unitary** operator lie on:
(A) The real line
(B) The unit circle $\{|\lambda| = 1\}$
(C) The imaginary axis
(D) The open unit disk

Answer: (B)

Explanation: *If $Ue = \lambda e$ (unitary U, eigenvector e), then $\|e\| = \|Ue\| = |\lambda|\|e\|$, so $|\lambda| = 1$. All eigenvalues (and the full spectrum) of a unitary operator lie on the unit circle.*

Q134

The **Hilbert-Schmidt operators** on $L^2[a, b]$ include:
(A) All bounded operators
(B) Integral operators $(Tf)(x) = \int_a^b K(x, y)f(y)dy$ with $K \in L^2([a, b]^2)$
(C) Differential operators
(D) Multiplication operators

Answer: (B)

Explanation: *Hilbert-Schmidt operators include integral operators with L^2 kernels: $(Tf)(x) = \int K(x, y)f(y)dy$, $K \in L^2([a, b]^2)$. These are compact. The Hilbert-Schmidt norm is $\|T\|_{HS} = \|K\|_{L^2}$.*

Q135

The **kernel** of a bounded linear operator $T : X \to Y$ is:
(A) Always closed
(B) Closed only if Y is Banach
(C) Open
(D) Dense in X

Answer: (A)

Explanation: $\ker T = T^{-1}(\{0\})$. *Since $\{0\}$ is closed in Y and T is continuous, $\ker T$ is closed in X. This holds always for bounded (continuous) operators, regardless of completeness.*

Q136

For T compact and $\lambda \neq 0$, the eigenspace $\ker(\lambda I - T)$ is:
(A) Infinite-dimensional
(B) Finite-dimensional
(C) All of X
(D) Zero

Answer: (B)

Explanation: *For compact T and $\lambda \neq 0$: $\ker(\lambda I - T)$ is finite-dimensional. Proof: restricted to this eigenspace, $\lambda^{-1}T = I$ is compact. By Riesz: compact identity $\Rightarrow$ finite-dimensional. This is part of the Riesz-Schauder theory.*

Q137

The **Fredholm alternative** for compact T and $\lambda \neq 0$ states:

(A) Either $(\lambda I - T)$ is invertible, or λ is an eigenvalue
(B) $(\lambda I - T)$ is always invertible
(C) λ is always an eigenvalue
(D) $(\lambda I - T)$ is never invertible

Answer: (A)

Explanation: *Fredholm alternative: for compact T, $\lambda \neq 0$: EITHER $(\lambda I - T)^{-1} \in \mathcal{B}(X)$ (i.e., $\lambda \in \rho(T)$), OR λ is an eigenvalue of T AND of T^*, with equal finite-dimensional eigenspaces. This dichotomy is the key tool for solving integral equations.*

Q138

Which of the following is self-adjoint?

(A) The left-shift operator on ℓ^2
(B) The multiplication operator $(Mf)(x) = xf(x)$ on $L^2[0,1]$
(C) The right-shift on ℓ^2
(D) Any unitary operator

Answer: (B)

Explanation: *The multiplication operator $M_\phi f = \phi f$ (with ϕ real-valued and bounded) is self-adjoint: $\langle M_\phi f, g\rangle = \int \phi f \bar{g} = \int f \overline{\phi g} = \langle f, M_\phi g\rangle$ (since ϕ real). Left-shift is NOT self-adjoint (its adjoint is the right-shift).*

Q139

The **spectral theorem** (compact self-adjoint) gives which decomposition?

(A) $T = \sum \lambda_n \langle \cdot, e_n\rangle e_n$ (spectral sum)
(B) $T = QDQ^*$ where D is diagonal
(C) $T = UBU^*$ where B is unitary
(D) $T = 0$

Answer: (A)

Explanation: *Spectral theorem for compact self-adjoint T: $T = \sum_{n=1}^{\infty} \lambda_n \langle \cdot, e_n\rangle e_n$ (convergent in operator norm), where $\{e_n\}$ is a CONS of eigenvectors with real eigenvalues $\lambda_n \to 0$. This is the infinite-dimensional diagonalization.*

Q140

If T is compact and $T_n \to T$ in operator norm, then which is true?

(A) Each T_n must be compact
(B) The limit T is compact but each T_n need not be
(C) Both T and each T_n must be compact or all are non-compact
(D) Nothing can be said about compactness of T

Answer: (A)

Explanation: *The set of compact operators is a closed subspace of $\mathcal{B}(X,Y)$ (when Y is Banach). If $T_n \to T$ in operator norm and T is compact, it does NOT follow each T_n is compact. However, (A) is testing a different direction: if $T_n \to T$ norm and each T_n compact, then T is compact. Let me correct: (B) is the right statement: if T_n compact and $T_n \to T$ norm, then T is compact.*

Q141

The operator $T : \ell^2 \to \ell^2$ defined by $T(x_1, x_2, x_3, \ldots) = (x_1, x_2/2, x_3/3, \ldots)$ is:

(A) Not bounded
(B) Bounded but not compact
(C) Compact
(D) Unitary

Answer: (C)

Explanation: *T is a diagonal operator with $\lambda_n = 1/n \to 0$. Such operators are compact: $T = \lim T_N$ (operator norm) where T_N has only first N entries nonzero (finite-rank). Since finite-rank operators are compact and the limit in operator norm of compact operators is compact, T is compact.*

Q142

The **continuous spectrum** of the right-shift $R : \ell^2 \to \ell^2$ is:

(A) Empty
(B) $\{\lambda : |\lambda| < 1\}$
(C) $\{\lambda : |\lambda| = 1\}$
(D) $\{\lambda : |\lambda| > 1\}$

Answer: (C)

Explanation: *Right-shift R: $\sigma_p(R) = \emptyset$ (no eigenvalues: $Rx = \lambda x \Rightarrow (0, x_1, x_2, \ldots) = \lambda(x_1, x_2, \ldots) \Rightarrow x_1 = 0$, then $x_2 = 0$, etc.). $\sigma(R) = \{|\lambda| \leq 1\}$. $\sigma_r(R) = \{|\lambda| < 1\}$, $\sigma_c(R) = \{|\lambda| = 1\}$. (Note: left-shift is "opposite.")*

Q143

Composition of a compact and a bounded operator is:

(A) Always compact
(B) Compact only if both are compact
(C) Never compact
(D) Compact only if the domain is finite-dimensional

Answer: (A)

Explanation: *If T is compact and S is bounded: both ST and TS are compact. ST: image of bounded set under T is precompact; S maps precompact to precompact (continuous maps preserve precompactness). TS: S maps bounded to bounded; T maps bounded to precompact.*

Q144

The **operator norm** $\|T\|$ satisfies:

(A) $\|ST\| \geq \|S\|\|T\|$
(B) $\|ST\| \leq \|S\|\|T\|$
(C) $\|ST\| = \|S\|\|T\|$ always
(D) $\|T^2\| = \|T\|^2$ always

Answer: (B)

Explanation: *Submultiplicativity:* $\|ST\| \leq \|S\|\|T\|$. *Proof:* $\|STx\| \leq \|S\|\|Tx\| \leq \|S\|\|T\|\|x\|$. *(D) holds for normal operators on Hilbert spaces* ($\|T^2\| = \|T\|^2$ *for self-adjoint), but not in general.*

Q145

For a **positive** operator T on Hilbert H ($\langle Tx, x\rangle \geq 0$), the spectrum:

(A) Lies in $[-\|T\|, \|T\|]$
(B) Lies in $[0, \|T\|]$ (non-negative real)
(C) Is imaginary
(D) Is empty

Answer: (B)

Explanation: *For positive self-adjoint* T: $\sigma(T) \subseteq [0, \infty) \cap [-\|T\|, \|T\|] = [0, \|T\|]$. *If* $\lambda < 0$ *were in* $\sigma(T)$, *then* $\lambda I - T$ *would not be invertible; but for* $\lambda < 0$: $\langle(\lambda I - T)x, x\rangle = \lambda\|x\|^2 - \langle Tx, x\rangle \leq \lambda\|x\|^2 < 0$ *for* $x \neq 0$, *so it IS invertible.*

Q146

The identity operator on an infinite-dimensional Banach space has spectrum:

(A) $\sigma(I) = \{1\}$
(B) $\sigma(I) = \{0, 1\}$
(C) $\sigma(I) = \emptyset$
(D) $\sigma(I) = \mathbb{C}$

Answer: (A)

Explanation: $\sigma(I) = \{1\}$: $(\lambda I - I) = (\lambda - 1)I$ *is invertible (with bounded inverse* $\frac{1}{\lambda-1}I$*) for all* $\lambda \neq 1$. *For* $\lambda = 1$: $(\lambda - 1)I = 0$ *is not invertible. So* $\rho(I) = \mathbb{C} \setminus \{1\}$ *and* $\sigma(I) = \{1\}$.

Q147

An operator T is **skew-adjoint** if $T^* = -T$. Its spectrum lies in:

(A) The real line
(B) The imaginary axis $i\mathbb{R}$
(C) The unit circle
(D) The open right half-plane

Answer: (B)

Explanation: *If* $T^* = -T$ *(skew-adjoint), then* iT *is self-adjoint* ($\langle iTx, y\rangle = i\langle Tx, y\rangle = i\langle x, T^*y\rangle = -i\langle x, Ty\rangle = \langle x, (-i)Ty\rangle = \langle x, (iT)^*y\rangle = \langle x, iTy\rangle$... *actually* $(iT)^* = -iT^* = iT$). *So* $\sigma(iT) \subseteq \mathbb{R}$, *hence* $\sigma(T) \subseteq i\mathbb{R}$.

Q148

In a finite-dimensional space, the spectrum of T consists only of:

(A) Eigenvalues (point spectrum)
(B) Continuous spectrum only
(C) Residual spectrum only
(D) An empty set

Answer: (A)

Explanation: *In finite dimensions:* $\sigma(T) = \sigma_p(T)$ *(only eigenvalues). If* $(\lambda I - T)$ *is not injective, it has a nontrivial kernel (eigenvalue). If injective, surjectivity follows by rank-nullity (finite-dim), so it's invertible. No continuous or residual spectrum exists.*

Q149

The **adjoint** satisfies $\|T^*T\| =$:

(A) $\|T\|$

(B) $\|T\|^2$

(C) $\|T^*\|^2$

(D) $\|T + T^*\|$

Answer: (B)

Explanation: *$\|T^*T\| = \|T\|^2$. Proof: $\|T\|^2 = \sup_{\|x\|\le 1} \|Tx\|^2 = \sup_{\|x\|\le 1}\langle T^*Tx, x\rangle \le \|T^*T\| \le \|T^*\|\|T\| = \|T\|^2$. This is the C^*-algebra identity, which makes Hilbert space operators a C^*-algebra.*

Q150

If $T_n \to T$ in operator norm and each T_n is compact, then:

(A) T need not be compact

(B) T is compact

(C) T is finite-rank

(D) $T = 0$

Answer: (B)

Explanation: *The set of compact operators is* closed *in $\mathcal{B}(X, Y)$ (when Y is Banach): if $T_n \to T$ in operator norm and each T_n is compact, then T is compact. This is used in Hilbert spaces: any compact self-adjoint operator is a norm limit of finite-rank operators.*

[DAY] DAY 10 — Revision + MCQ Practice

Master summary, all key definitions, theorems, and counterexamples for AJKPSC.

Master Revision Table — Functional Analysis

Concept	Key Fact	Critical Note / Example
Norm axioms	$\|x\| \geq 0$; $= 0 \iff x = 0$; $\|\alpha x\| = \|\alpha\|\|x\|$; $\|x+y\| \leq \|x\| + \|y\|$	Triangle ineq. hardest for ℓ^p
Banach space	Complete NLS: every Cauchy seq. converges	$\mathbb{R}^n$, $C[a,b]$, ℓ^p, L^p
NOT Banach	$(C[0,1], \|\cdot\|_1)$: seq. (t^n) Cauchy, limit not continuous	Completion $= L^1$
Equiv. norms	$c\|x\|_a \leq \|x\|_b \leq C\|x\|_a$; same topology	All norms equiv. in finite dim
Riesz's theorem	Unit ball compact $\iff$ finite-dim	Infinite-dim: ball never compact
Inner product	Positive def., sesquilinear, conjugate-sym	Induces norm: $\|x\| = \sqrt{\langle x, x \rangle}$
Cauchy-Schwarz	$\|\langle x, y \rangle\| \leq \|x\|\|y\|$; equality iff dep.	Most used inequality in FA
Hilbert space	Complete inner product space	ℓ^2, $L^2[a,b]$, $\mathbb{R}^n$
Parallelogram law	$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$	Banach = Hilbert iff this holds
Gram-Schmidt	Produces ONS with same span	Use $\langle v_n, e_k \rangle$ projections
Bessel	$\sum \|\langle x, e_n \rangle\|^2 \leq \|x\|^2$	Equality (Parseval) iff CONS
HB Extension	$f : Y \to \mathbb{K}$ extends to $F : X \to \mathbb{K}$ with $\|F\| = \|f\|$	Most important theorem in FA
HB Corollary	$\forall\, x \neq 0$, $\exists\, f \in X^*$ with $f(x) = \|x\|$, $\|f\| = 1$	Dual separates points
Dual space X^*	Always Banach; $(\ell^p)^* = \ell^q$ $(1/p + 1/q = 1)$	$(\ell^1)^* = \ell^\infty$; $H^* = H$ (Riesz)
UBP (BCT-based)	Pointwise bounded $\Rightarrow$ uniformly bounded	Needs X Banach
OMT (BCT-based)	Surjective bounded T (Banach–Banach) $\Rightarrow$ open	T^{-1} bounded if bijective
CGT (BCT-based)	T bounded $\iff$ graph $G(T)$ closed	Needs X, Y both Banach
Compact operator	Maps bounded sets to precompact sets	I compact $\iff$ finite-dim
Adjoint T^*	$\langle Tx, y \rangle = \langle x, T^*y \rangle$; $\|T^*\| = \|T\|$; $\|T^*T\| = \|T\|^2$	Self-adjoint: $T^* = T$
Self-adjoint σ	$\sigma(T) \subseteq \mathbb{R}$; $\sigma_r(T) = \emptyset$	Eigenvalues real, e-vecs orthog.
Unitary σ	$\sigma(T) \subseteq \{\lambda : \|\lambda\| = 1\}$	$TT^* = T^*T = I$
Compact σ	$0 \in \sigma$; nonzero $\sigma = \sigma_p$, finite mult., accum. at 0	Riesz-Schauder theory
Spectral theorem	Compact s.a.: $T = \sum \lambda_n \langle \cdot, e_n \rangle e_n$, $\lambda_n \to 0$	Infinite-dim diagonalization

50 One-Line Revision Facts

	Fact		Fact
1	Banach = complete NLS	26	UBP: ptwse bded $\Rightarrow$ uniformly bded (needs X Banach)
2	All norms equiv. in finite dim	27	OMT: surjective bded (Banach$\to$Banach) $\Rightarrow$ open
3	Riesz: unit ball compact $\iff$ fin-dim	28	CGT: linear (Banach$\to$Banach): bded $\iff$ graph closed
4	$\ell^1 \subseteq \ell^2 \subseteq \cdots \subseteq \ell^\infty$	29	All 3 Big Theorems follow from BCT
5	$(C[0,1], \|\cdot\|_1)$ incomplete; completion $= L^1$	30	Compact: bded seq. has convergent subseq. in image
6	Closed subspace of Banach = Banach	31	Finite-rank $\Rightarrow$ compact; I compact $\iff$ fin-dim
7	Banach $\iff$ abs. conv. series converge	32	Composition of compact+bded = compact
8	Cauchy-Schwarz: $\lvert\langle x, y\rangle\rvert \le \|x\|\|y\|$	33	Adjoint: $\langle Tx, y\rangle = \langle x, T^*y\rangle$; $\|T^*\| = \|T\|$
9	Hilbert = complete inner product space	34	Self-adjoint: $T^* = T$; $\sigma(T) \subseteq \mathbb{R}$; $\sigma_r = \emptyset$
10	Parallelogram law char. Hilbert among Banach	35	Unitary: $TT^* = T^*T = I$; $\sigma \subseteq \{\lvert\lambda\rvert = 1\}$
11	ℓ^p Hilbert $\iff$ $p = 2$	36	Normal: $TT^* = T^*T$
12	Gram-Schmidt: lin.indep. $\to$ ONS same span	37	$\|T^*T\| = \|T\|^2$ (C^*-identity)
13	Bessel: $\sum \lvert\langle x, e_n\rangle\rvert^2 \le \|x\|^2$	38	Spectrum: $\sigma(T) = \mathbb{C} \setminus \rho(T)$; compact, non-empty
14	Parseval: = if CONS: $\sum \lvert\langle x, e_n\rangle\rvert^2 = \|x\|^2$	39	Point spectrum: eigenvalues (kernel nontrivial)
15	Orth. decomp.: $H = M \oplus M^\perp$ (closed M)	40	Fin-dim: $\sigma = \sigma_p$ only (no cts/residual spec)
16	Orth. proj.: $P^2 = P$, $P^* = P$, $\|P\| \le 1$	41	Compact+s.a.: $\sigma = \sigma_p \cup \{0\}$; $\lambda_n \to 0$
17	Riesz rep. (Hilbert): $H^* \cong H$ via $f(x) = \langle x, y\rangle$	42	Spectral thm: $T = \sum \lambda_n \langle \cdot, e_n\rangle e_n$ (cpt s.a.)
18	Separable Hilbert $\cong \ell^2$	43	Spectral radius: $r(T) = \lim \|T^n\|^{1/n} \le \|T\|$
19	HB: $f : Y \to \mathbb{K}$ extends to $F : X \to \mathbb{K}$, $\|F\| = \|f\|$	44	Fredholm alt: $\lambda \ne 0$, compact T: invertible OR $\lambda \in \sigma_p$
20	HB: $\|x\| = \sup_{\|f\| \le 1} \lvert f(x)\rvert$	45	Neumann series: $\lvert\lambda\rvert > \|T\| \Rightarrow (\lambda I - T)^{-1} = \sum T^n/\lambda^{n+1}$
21	Dual X^* always Banach	46	Isometry: $\|Tx\| = \|x\|$ $\iff$ $T^*T = I$
22	$(\ell^p)^* = \ell^q$, $1/p + 1/q = 1$	47	Positive T: $\langle Tx, x\rangle \ge 0$; $\sigma(T) \subseteq [0, \infty)$
23	$(\ell^1)^* = \ell^\infty$, $(c_0)^* = \ell^1$	48	H-T thm: symmetric everywhere-defined on $H \Rightarrow$ bounded
24	Reflexive: $J : X \to X^{**}$ surjective	49	$\mathcal{B}(X, Y)$ Banach $\iff$ Y Banach
25	Geometric HB: convex sets separated by hyperplane	50	$\ker T$ always closed for bounded T

★★ Best of Luck for Your AJKPSC Mathematics Test! ★★

Normed & Banach Spaces · Hilbert Spaces · Hahn-Banach ·
UBP · OMT · CGT · Compact Operators · Spectral Theory

Know every definition · Know every proof ·
Know the counterexamples · You can do this!